Open-book reference: DSP L01–L10 (full course)

Cheat sheet · Use from Sat 14 Nov · pack REFD

Covers every lecture, L01 to L10, and replaces the weeks 1–4 strip of the earlier version. Notation follows the course: uppercase F, Ω for continuous time (CT); lowercase f, ω for discrete time (DT). $\omega = 2\pi F/F_s = \Omega T$ (rad/sample). The course defines **sinc** $x = \sin x/x$ (MATLAB/numpy use $\sin(\pi x)/(\pi x)$, T9 Q1). δ_p = **minimum** passband gain, δ_s = maximum stopband gain (linear). "↑" marks $n = 0$. Sample Test 1 (50 min, 20 marks, open book) covered L02–L04; L05–L10 have never been tested.

L01 Overview: why DSP, and signal types

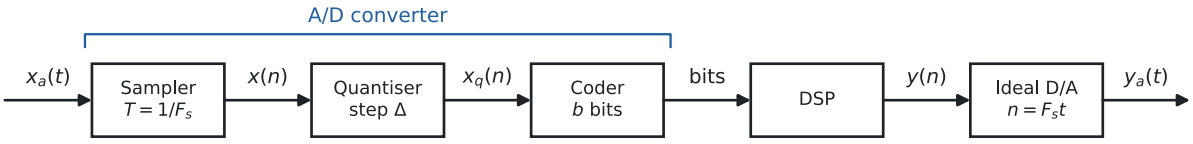
Advantage of digital (slide 6)	Why (analogue vs digital filter case study, slides 7–8)
Flexibility (programmable)	Analogue: a new cutoff needs a new R_1 and/or C_1; new filter type means a new circuit. Digital: just change $h(n)$
Accuracy, precision	DSP computes $y(n) = \sum_k h(k)x(n-k)$ with multiply-accumulate; no component tolerance
Low-cost implementation	A long $h(n)$ (high order) is cheap in software; high-order analogue filters are expensive
Reliability	No drift or failure with time and temperature (analogue R_1, C_1 drift)

Classification (slides 9–11)	Meaning
Continuous-time / discrete-time	$x(t)$, $t \in \mathbb{R}$ / $x(n)$, $n \in \mathbb{Z}$. DT usually from sampling; some are inherently DT (stock prices)
Continuous-valued / discrete-valued	amplitude real / amplitude in a finite set Q (quantised)
Digital	discrete-time **and** discrete-valued
Deterministic / random	explicit formula (e.g. $5e^{-j50\pi t}$) / described by a pdf (e.g. Gaussian, mean μ, variance σ^2)

L02 Sampling and DT signals

CT vs DT frequency (slides 4–7)

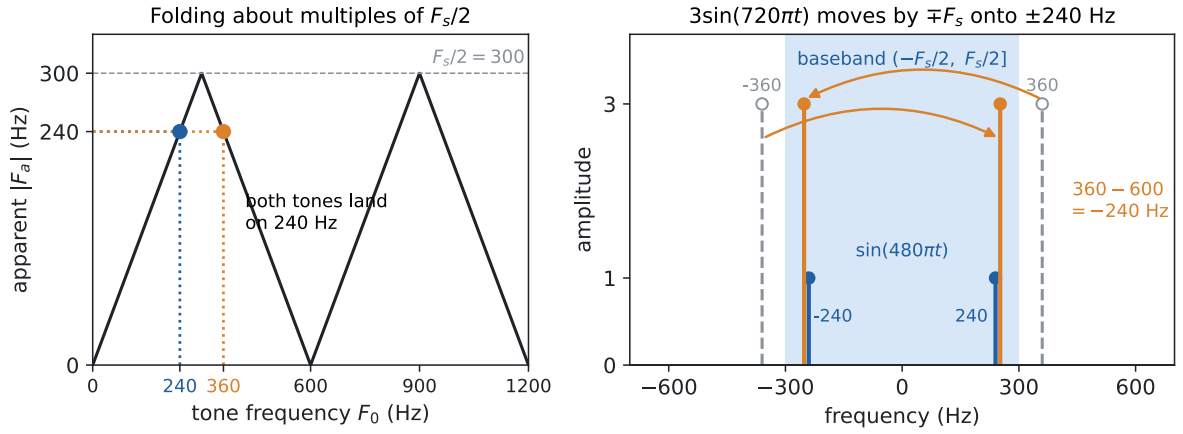
	CT $x(t) = A\cos(\Omega_0 t + \phi)$	DT $x(n) = A\cos(\omega_0 n + \phi)$
Units	$\Omega_0 = 2\pi F_0$ rad/s, F_0 Hz	$\omega_0 = 2\pi f_0$ rad (per sample), f_0 cycles/sample
Periodic?	always, $T_0 = 1/\lvert F_0\rvert = 2\pi/\lvert\Omega_0\rvert$	only if $f_0 = k/N$ is rational; N = denominator in lowest terms
Uniqueness	every F_0 gives a different signal	ω_0 and $\omega_0 + 2\pi k$ are the same signal: unique only on $-\pi < \omega_0 \le \pi$ ($-\frac{1}{2} < f_0 \le \frac{1}{2}$)
Fastest	none	$\omega = \pm\pi$: $\cos(\pi n) = (-1)^n$



L02 signal chain: A/D (sampler, quantiser, coder), DSP, D/A

Sampling, aliasing, reconstruction (slides 9–10)

Item	Formula	Note
Sampler	$x(n) = x_a(nT), T = 1/F_s$	$f_0 = F_0/F_s, \omega_0 = \Omega_0/F_s = 2\pi F_0/F_s$
Highest representable	$F_{max} = F_s/2, \Omega_{max} = \pi F_s$	$F_s/2$ = folding frequency
Sampling theorem	exact recovery by ideal LPF iff $F_s > 2F_{max}$	$2F_{max}$ = **Nyquist rate** (bandpass sampling is the exception, L11)
Apparent (alias) frequency	$F_a = F_0 - kF_s$, k integer, so $-F_s/2 < F_a \leq F_s/2$	same as reducing ω_0 into $(-\pi, \pi]$
Sign after aliasing	$\cos(-\theta) = \cos\theta, \sin(-\theta) = -\sin\theta$	a sine aliased to negative F_a flips sign (T1 Q2)
Integer f_0	$\cos(2\pi kn) = 1, \cos(\pi n) = (-1)^n$	ST1 Q1(a): 200 Hz at $F_s = 100$ Hz is DC
Ideal DAC	$y(t) = x(n)\|_{n=F_s t}$ **after** reducing ω	$y(t) \neq x_a(t)$ whenever aliasing occurred



Aliasing, T1 Q2: 240 Hz and 360 Hz sampled at 600 Hz both appear at 240 Hz; the aliased sine flips sign

Quantiser, coder, DAC (slides 11–16)

Item	Formula	Note
Quantiser	$x_q(n) = Q(x(n))$, $e_q(n) = x_q(n) - x(n)$	rounding: $-\Delta/2 \le e_q \le \Delta/2$
Step size	$\Delta = \dfrac{x_{max} - x_{min}}{L-1}$	$L-1$, not L; range of $A\cos$ is $2A$
Levels from Δ	$L = \dfrac{x_{max} - x_{min}}{\Delta} + 1$	
Bits	$2^b \ge L$, $b = \lceil \log_2 L \rceil$	round **up** (636 levels: 9.31 → 10 bits)
From b	$L = 2^b$, $\Delta = \text{range}/(2^b - 1)$	8 bits, 1 V: $\Delta = 1/255 = 3.92$ mV
Bit rate	bF_s bits/s	max recoverable frequency $F_s/2$
DAC	interpolates between samples: zero-order hold (most popular), linear, higher-order	approximates the ideal LPF
ZOH	holds each sample for T; $\lvert H_{ZOH}(F)\rvert = T\left\lvert \frac{\sin(\pi FT)}{\pi FT}\right\rvert$ (standard result; slide 16 is a figure)	droop: $2/\pi = -3.92$ dB at $F_s/2$; corrected by an analogue **inverse-sinc** post-filter that also removes images

DT signals (slides 18–24)

Item	Definition / rule	Note
Sequence form	$\{-1, 0, \underset{\uparrow}{3}, -2, 1\}$; all other samples 0	always mark $n = 0$
$\delta(n)$, $u(n)$, $u_r(n)$	1 at $n = 0$; 1 for $n \ge 0$; n for $n \ge 0$	$\delta(n) = u(n) - u(n-1)$; $u(n) = \sum_{k \ge 0} \delta(n-k)$
Windows	$u(n+a) - u(n-b) = 1$ for $-a \le n \le b-1$	$u(n) - u(n-N)$: N ones from 0
Sifting	$x(n)\delta(n-k) = x(k)\delta(n-k)$; $x(n) = \sum_k x(k)\delta(n-k)$	
Exponential	a^n; $a = re^{j\theta}$: $\lvert x\rvert = r^n$, $\angle x = \theta n$	course writes $(1/2)^n$ as 2^{-n}
Energy	$E_\infty = \sum \lvert x(n)\rvert^2$; energy signal if $0 < E < \infty$ (then $P = 0$)	finite length is always energy; $a^n u(n)$, $\lvert a\rvert < 1$: $E = 1/(1 - \lvert a\rvert^2)$
Power	$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{-N}^{N} \lvert x\rvert^2$; power signal if $0 < P < \infty$ (then $E = \infty$)	periodic: $P = \frac{1}{N}\sum_0^{N-1} \lvert x\rvert^2$; $u(n)$: $P = \frac{1}{2}$; $nu(n)$: neither
DT periodic	$x(n) = x(n+N)$ for all n; smallest N = fundamental period	
Sums/products of sinusoids	each $f_i = k_i/N_i$ (lowest terms); $N = \text{LCM}(N_i)$	one irrational f_i (a π left over) makes it aperiodic
Even / odd (real x)	even $x(n) = x(-n)$; odd $x(-n) = -x(n)$, so $x(0) = 0$	can be neither ($nu(n)$)
Decomposition	$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$, $x_o(n) = \frac{1}{2}[x(n) - x(-n)]$, $x = x_e + x_o$	$\{\underset{\uparrow}{1}, 2, 3\}$: $x_e = \{1.5, 1, \underset{\uparrow}{1}, 1, 1.5\}$, $x_o = \{-1.5, -1, \underset{\uparrow}{0}, 1, 1.5\}$

Sequence operations (slides 23–24)

Operation	Effect	Exam method
Shift $x(n-n_0)$	$n_0>0$: delay (moves **right**); $n_0<0$: advance (left)	add n_0 to every index
Reversal $x(-n)$	mirror about $n=0$	negate every index
Downsample $x(Mn)$	keep indices that are multiples of M, divide them by M	samples are lost
Upsample $x(n/L)$	index $m \to Lm$, zeros between	nonzero only at multiples of L
$(-1)^n x(n)$	flip the sign of odd-index samples	modulation by $\omega=\pi$
Amplitude, sum, product	$Ax(n)$, x_1+x_2, x_1x_2 sample by sample	line up the index rows
Order rule $x(\alpha n+\beta)$	first shift: $v(n)=x(n+\beta)$ (advance if $\beta>0$); then scale $v(\alpha n)$: compress by $\lvert\alpha\rvert>1$, stretch by $1/\lvert\alpha\rvert$ if $\lvert\alpha\rvert<1$, reverse if $\alpha<0$	safest: for each n evaluate the index $\alpha n+\beta$ and look up x

L03 DT systems, LTI, correlation

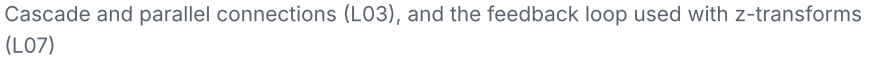
$y(n)=T\{x(n)\}$. To **prove** a property give a general argument; to **disprove** it one concrete counterexample is enough.

Property	Test	Standard counterexample (fails)	Passes, e.g.
Static (memoryless)	$y(n)$ uses only $x(n)$ at the same n	$y(n)=x(n-1)$: $y(1)=x(0)$; any sum, $y(n-k)$, $x(2n)$, $x(-n)$	$x^2(n)$, $x(n)\cos(0.5\pi n-0.1\pi)$
Linear	$T\{a_1x_1+a_2x_2\}=a_1T\{x_1\}+a_2T\{x_2\}$	$y=x^2$: $2x\to 4y$. $y=1+2x$: $x=0\to y=1\neq 0$. $\cos x$, $\lvert x\rvert$, e^x	$5x(n)+x(n+1)$, $x(n)\cos(0.8\pi n)$, $nx(n)$
Time-invariant	feed $x(n-d)$ (leave explicit n alone) and compare with $y(n-d)$ (every $n\to n-d$)	$y=nx(n)$: $\delta(n)\to 0$ but $\delta(n-1)\to\delta(n-1)$. $y=x(2n)$: $\delta(n-1)\to 0$, not $\delta(n-1)$ (T3 Q1.4)	$2x(n)-3x(n-2)$, $x^3(n)$
Causal	$y(n)$ uses no $x(n+k)$, $k>0$, at **any** n	$y=x(n+1)$; $x(-n)$ at $n=-1$ needs $x(1)$; $x(2n)$ at $n=1$ needs $x(2)$; centred average	$5x(n)+x(n-1)$, $(n+5)x(n)$
BIBO stable	$\lvert x\rvert<M_x\Rightarrow\lvert y\rvert<M_y$	accumulator $\sum_{-\infty}^{n}x(k)$ with $x=u(n)$ gives $(n+1)\to\infty$; $nx(n)$ with $x=u(n)$	$x^3(n)$, $\cos x(n)$ (≤ 1)

Reading an LCCDE (ST1 Q3, T3 Q1). Constant term (-1) → nonlinear. Coefficient depending on n ($ny[n-1]$, $(n-1)x[n]$, $x[n-2]u[n]$) → time-varying, still linear. $\cos(0.2\pi)$ is a constant; $\cos(0.2\pi n)$ is not. Any $x[n+k]$ → non-causal. Any delayed term → dynamic. Constant coefficients and no offset → LTI.

LTI from observed pairs (T2 Q2). Use time invariance to shift a known pair, then test scaling/superposition against another pair; one mismatch proves nonlinear.

Connection (slides 11–12)	General system	LTI
Cascade	$T_c = T_2 \circ T_1$ (order matters in general)	$h = h_1 * h_2$, order irrelevant
Parallel	$T_p = T_1 + T_2$	$h = h_1 + h_2$ (minus sign at a subtracting junction)
Mixed (T2 Q4)	work inside out	$h = h_1 * (h_2 - h_3 * h_4)$



Cascade and parallel connections (L03), and the feedback loop used with z-transforms (L07)

Correlation (slides 15–18)

Item	Formula	Note
Cross (lecture)	$r_{xy}(l) = \sum_n x(n)y(n-l) = \sum_n x(n+l)y(n)$	like convolution **without the flip**
Auto	$r_{xx}(l) = \sum_n x(n)x(n-l)$	even; $r_{xx}(0) = E_x$ is the maximum
Properties	$r_{xy}(l) = r_{yx}(-l)$; $\lvert r_{xy}(l)\rvert \le \sqrt{E_x E_y}$	
Via convolution	$r_{xy}(l) = x(l) * y(-l)$	reverse one signal, convolve (long multiplication)
Tutorial form (T3 Q7)	$R_{xy}(l) = \sum_k x(k)y(k+l) = x(-n) * y(n) = r_{xy}(-l) = r_{yx}(l)$	lag-reversed: state which definition you use
Periodic, period N	$r_{xy}(l) = \frac{1}{N}\sum_{n=0}^{N-1} x(n)y(n-l)$	also periodic with N
Radar delay	echo $y(n) = \alpha x(n-d) + w(n)$; $d = \mathrm{round}(2F_s r/c)$; $\hat{d} = \arg\max_l \lvert r_{yx}(l)\rvert$	range $r = cd/(2F_s)$; resolution $c/(2F_s)$

T3 Q7 ($x = \{\underset{\uparrow}{-1}, 0, 1, 3\}$, $y = \{\underset{\uparrow}{-2}, 3, 1, 2, 4\}$): $R_{xy}(l) = \{-6, 7, 6, 9, 11, 3, -2, -4\}$ for $l = -3..4$. Lecture $r_{xy}(l)$ is the same list reversed, $l = -4..3$. Both definitions give a delay estimate of 1 (peak 11).

L04 LCCDEs, impulse response and convolution

$\sum_{p=0}^{k} a_p y(n-p) = \sum_{q=0}^{m} b_q x(n-q)$, $a_0 = 1$, order k. Homogeneous if the right side is 0. Recursive form (slide 13): $y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k)$ (move y terms right, **flip signs**).



Direct form I of the recursive form

Characteristic equation and modes (slides 5–7). Put $y = \lambda^n$ in the homogeneous equation: $\lambda^k + a_1\lambda^{k-1} + \cdots + a_k = 0$. Use the y **coefficients only**.

Roots	Homogeneous solution $y_h(n)$	Example
Distinct real $\lambda_1, \ldots, \lambda_k$	$\sum C_p \lambda_p^n$	T3 Q3: $\lambda^2 - \frac{3}{4}\lambda + \frac{1}{8}$, $\lambda = \frac{1}{2}, \frac{1}{4}$: $C_1 2^{-n} + C_2 4^{-n}$
Repeated λ (r times)	$(C_1 + C_2 n + \cdots + C_r n^{r-1})\lambda^n$	T3 Q4: $\lambda = -\frac{1}{2}$ twice: $C_1(-2)^{-n} + C_2 n(-2)^{-n}$; $\lambda = 1$ twice: $C_1 + C_2 n$ (T3 Q5c)
Complex pair $re^{\pm j\theta}$	$r^n(A\cos\theta n + B\sin\theta n)$	ST1 Q4: $\lambda^2 - \lambda + 1$, $\lambda = e^{\pm j\pi/3}$: $\cos\frac{\pi n}{3}$, $\sin\frac{\pi n}{3}$
No $y(n-k)$ terms	no modes ("the modes are zero / there are none"): FIR	T3 Q2
Stability	causal system stable iff all $\lvert\lambda_p\rvert < 1$	$\lvert\lambda\rvert = 1$ ("marginal") is **not** BIBO stable

Particular solution (slide 8): guess the input's form, substitute into the full LCCDE, cancel the common factor, solve for K.

Input $x(n)$	**Trial** $y_p(n)$
constant A	K
Aa^n (incl. $A3^{-n}$)	Ka^n; if a is a characteristic root, Kna^n
$A\cos\omega_0 n$ or $A\sin\omega_0 n$	$K_1\cos\omega_0 n + K_2\sin\omega_0 n$
An^M	$K_0 + K_1 n + \cdots + K_M n^M$
general	convolution, $y_{zs} = x * h$

Total solution with ICs. $y = y_h + y_p$. Get $y(0), \ldots, y(k-1)$ **by recursion** from the given $y(-1), y(-2)$ (or zeros if "at rest"), then solve the $k \times k$ system for the C_p. Never put $y(-1)$ into the closed form; it holds only for $n \geq 0$. Zero-input/zero-state view: $y = y_{zi} + y_{zs}$, y_{zi} from ICs with $x = 0$, $y_{zs} = x * h$ with zero ICs. With nonzero ICs the system is not LTI (zero input does not give zero output).

$h(n)$ from an LCCDE (slides 13–14). $h(n)$ = response to $x = \delta(n)$ with **zero ICs**.

Case	$h(n)$
No feedback (FIR, non-recursive)	$h(n) = b_n$: read off $\{b_0, b_1, \ldots, b_M\}$, including zeros (T3 Q5a: $\{1, 2, 3, 0, 4\}$)
Feedback, $m < k$	$h(n) = \sum C_p \lambda_p^n\, u(n)$; fit to $h(0), \ldots, h(k-1)$ from recursion
Feedback, $m \geq k$	modes hold only from $n = m - k + 1$: add $\sum_{i=0}^{m-k} D_i \delta(n-i)$ and fit the extra samples
Alternative	find h_1 for right side $\delta(n)$ only, then $h(n) = \sum_q b_q h_1(n-q)$ (linearity + TI)

Convolution (slide 9). $y(n) = \sum_k x(k)h(n-k) = \sum_k h(k)x(n-k)$. Commutative, associative, distributive; $x * \delta(n-d) = x(n-d)$; $x * u = \sum_{-\infty}^{n} x(k)$.

Method	Rule	Check
Tabular (finite)	write h on top; rows $x(0)h$, $x(1)h$ shifted one place right, …; add columns (no carries)	start index = sum of start indices; length $N_x + N_h - 1$; $\sum y = (\sum x)(\sum h)$
Closed form	keep the steps; $u(k) \Rightarrow k \geq 0$, $u(n-k) \Rightarrow k \leq n$; these give the limits **and** the n range; factor out k-free terms; geometric sum	$y(0) = h(0)x(0)$, $y(1) = h(0)x(1) + h(1)x(0)$
Finite window (T2 Q3)	slide the flipped window; one case per overlap stage (entering, full, leaving)	limits $k \in [\max(\cdot), \min(\cdot)]$

Standard result (all $\times u(n)$)	
$a^n u(n) * b^n u(n)$, $a \neq b$	$\dfrac{a^{n+1} - b^{n+1}}{a-b} u(n)$
$a^n u(n) * a^n u(n)$	$(n+1)a^n u(n)$
$a^n u(n) * u(n)$	$\dfrac{1 - a^{n+1}}{1-a} u(n)$
$u(n) * u(n)$	$(n+1)u(n)$
$e^{j\omega_0 n} u(n) * 2^{-n} u(n)$ (T3 Q6c)	$e^{j\omega_0 n} \dfrac{1 - (2e^{j\omega_0})^{-(n+1)}}{1 - (2e^{j\omega_0})^{-1}} u(n)$; cosine input: real part (T3 Q6d), since h is real

Properties read off $h(n)$ (slides 10–11)

Property	Condition on $h(n)$	Note
Static	$h(n) = K\delta(n)$	
Causal	$h(n) = 0$ for $n < 0$	
Stable	$\sum \lvert h(k) \rvert < \infty$	$\sum \lvert a \rvert^n < \infty$ iff $\lvert a \rvert < 1$; $u(n)$, $(n+1)u(n)$, $\cos(\omega_0 n)u(n)$ are **not** summable
FIR	finitely many nonzero samples	no feedback; always stable; implement by the convolution sum
IIR	infinite length	recursive (feedback) implementation; convolution sum cannot be computed directly

Geometric series box

Sum	Value
$\sum_{k=0}^{N-1} a^k$	$\frac{1-a^N}{1-a}$ ($a \neq 1$); N if $a = 1$
$\sum_{k=0}^{n} a^k$	$\frac{1-a^{n+1}}{1-a}$ ($n+1$ terms)
$\sum_{k=M}^{N} a^k$	$\frac{a^M - a^{N+1}}{1-a}$; number of terms $N - M + 1$
$\sum_{k=0}^{\infty} a^k$	$\frac{1}{1-a}$, $\lvert a\rvert < 1$
$\sum_{k=M}^{\infty} a^k$	$\frac{a^M}{1-a}$, $\lvert a\rvert < 1$
$\sum_{k=0}^{\infty} k a^k$	$\frac{a}{(1-a)^2}$, $\lvert a\rvert < 1$
$\sum_{k=0}^{N-1} k$	$\frac{N(N-1)}{2}$
Euler	$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$, $e^{j\theta} = \cos\theta + j\sin\theta$
Pull-out trick	$\sum_{k=0}^{n} b^k a^{n-k} = a^n \sum_{k=0}^{n} (b/a)^k$

L05 DTFT, pairs and properties

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}, \qquad x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n}\, d\omega$$

Exists if $\sum |x| < \infty$ or $\sum |x|^2 < \infty$. Period 2π. DTFS: $c_k = \frac{1}{N} \sum_{\langle N \rangle} x(n) e^{-jk\omega_0 n}$, $\omega_0 = 2\pi/N$, $c_{k+N} = c_k$, power $= \sum |c_k|^2$.

DTFT pairs

$x(n)$	$X(\omega)$	Condition
$\delta(n-n_0)$	$e^{-j\omega n_0}$	
$a^n u(n)$	$\dfrac{1}{1-ae^{-j\omega}}$	$\lvert a\rvert<1$
$b^n u(-n)$	$\dfrac{1}{1-b^{-1}e^{j\omega}}$	$\lvert b\rvert>1$
$na^n u(n)$	$\dfrac{ae^{-j\omega}}{(1-ae^{-j\omega})^2}$	$\lvert a\rvert<1$
$(n+1)a^n u(n)$	$\dfrac{1}{(1-ae^{-j\omega})^2}$	$\lvert a\rvert<1$
$a^{\lvert n\rvert}$	$\dfrac{1-a^2}{1-2a\cos\omega+a^2}$	$\lvert a\rvert<1$
$u(n)-u(n-M)$ (M ones from 0)	$\dfrac{1-e^{-j\omega M}}{1-e^{-j\omega}}=e^{-j\omega(M-1)/2}\dfrac{\sin(M\omega/2)}{\sin(\omega/2)}$	
1 for $-L\le n\le L$	$\dfrac{\sin((2L+1)\omega/2)}{\sin(\omega/2)}$	real, even
$\dfrac{\sin\omega_c n}{\pi n}$ (value ω_c/π at $n=0$)	1 for $\lvert\omega\rvert<\omega_c$, else 0	ideal LPF (T9 Q1)
$e^{j\omega_0 n}$	$2\pi\delta(\omega-\omega_0)$	in $(-\pi,\pi]$, repeats every 2π
$\cos\omega_0 n$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	in $(-\pi,\pi]$
$u(n)$	$\dfrac{1}{1-e^{-j\omega}}+\pi\sum_k\delta(\omega-2\pi k)$	from accumulation

DTFT properties (L05 slides 21–26)

Property	Time	Frequency
Shift	$x(n-n_0)$	$e^{-j\omega n_0}X(\omega)$
Modulation	$e^{j\omega_0 n}x(n)$; $(-1)^n x(n)$	$X(\omega-\omega_0)$; $X(\omega-\pi)$
Conjugate	$x^*(n)$	$X^*(-\omega)$. Real x: $\lvert X\rvert$ even, θ odd
Reversal	$x(-n)$	$X(-\omega)$
Difference	$x(n)-x(n-1)$	$(1-e^{-j\omega})X(\omega)$
Accumulation	$\sum_{k=-\infty}^{n}x(k)$	$\frac{X(\omega)}{1-e^{-j\omega}}+\pi X(0)\sum_k\delta(\omega-2\pi k)$
Time expansion	$x_{(k)}(n)$ (insert $k-1$ zeros)	$X(k\omega)$: compresses, images at $2\pi/k$
Freq. derivative	$n\,x(n)$	$j\,dX/d\omega$
Convolution	$x*h$	$X(\omega)H(\omega)$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi}\int_{2\pi}X_1(\lambda)X_2(\omega-\lambda)d\lambda$
Parseval	$\sum\lvert x\rvert^2$	$\frac{1}{2\pi}\int_{2\pi}\lvert X\rvert^2 d\omega$
Sampling	$x(n)=x_a(nT)$	$X(\omega)=\frac{1}{T}\sum_k X_a((\omega-2\pi k)F_s)$; recover with LPF gain T

Shortcuts without computing $X(\omega)$ (T4 Q2): $X(0)=\sum x(n)$; $X(\pi)=\sum(-1)^n x(n)$; $\int_{-\pi}^{\pi}X\,d\omega=2\pi x(0)$; $\int_{-\pi}^{\pi}\lvert X\rvert^2 d\omega=2\pi\sum\lvert x\rvert^2$; $\int_{-\pi}^{\pi}X(\omega)e^{j\omega n_0}d\omega=2\pi x(n_0)$;

$X'(0) = -j\sum nx(n)$; $\angle$ of a negative real number $= \pm\pi$. Shifted one-sided: $a^n u(n+n_0) \to \dfrac{a^{-n_0}e^{j\omega n_0}}{1-ae^{-j\omega}}$ (T4 Q1c: $81e^{j4\omega}/(1-\frac{1}{3}e^{-j\omega})$).

Frequency response from an LCCDE and from poles and zeros (L05 slides 27–35)

Item	Formula	Trap
Definition	$H(\omega) = \sum h(n)e^{-j\omega n} = H(z)\|_{z=e^{j\omega}}$	only if ROC contains $\|z\| = 1$ (stable)
From LCCDE $y(n) = -\sum a_k y(n-k) + \sum b_k x(n-k)$	$H(\omega) = \dfrac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{1+\sum_{k=1}^{N} a_k e^{-jk\omega}}$	move all y terms left first: signs of a_k flip
Eigen-response	$e^{j\omega n} \to H(\omega)e^{j\omega n}$; $A\cos(\omega n+\phi) \to A\|H(\omega)\|\cos(\omega n+\phi+\theta(\omega))$	keep input phase ϕ; sin stays sin
No new frequencies	$Y = HX$	a new frequency proves **not** LTI; the reverse proves nothing
First order	$\left\|\dfrac{1}{1-ae^{-j\omega}}\right\| = \dfrac{1}{\sqrt{1-2a\cos\omega+a^2}}$	
Symmetric FIR	factor out the middle delay: $1+e^{-j2L\omega} = 2e^{-jL\omega}\cos L\omega$	negative real amplitude: add π to phase
Pole-zero form	$H(\omega) = b_0 e^{j\omega(N-M)}\dfrac{\prod(e^{j\omega}-z_k)}{\prod(e^{j\omega}-p_k)}$	vectors **from** the zero/pole **to** $e^{j\omega}$
Magnitude	$\|H\| = \|b_0\|\dfrac{\prod V_k(\omega)}{\prod U_k(\omega)}$ (V: zero distances, U: pole distances)	poles/zeros at origin don't change $\|H\|$
Phase	$\theta = \angle b_0 + \omega(N-M) + \sum\angle(e^{j\omega}-z_k) - \sum\angle(e^{j\omega}-p_k)$	origin terms give linear phase only
Sketch	walk $e^{j\omega}$ from $z=1$ ($\omega=0$) to $z=-1$ ($\omega=\pi$)	zero on circle: $\|H\|=0$; pole near circle: peak

L06 DFT

$$X(k) = \sum_{n=0}^{N-1} x(n)W_N^{kn}, \quad x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)W_N^{-kn}, \quad W_N = e^{-j2\pi/N}$$

$\mathbf{X} = \mathbf{W}_N\mathbf{x}$; $\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$, $\mathbf{W}_N^{-1} = \frac{1}{N}\mathbf{W}_N^*$. $X(k) = X(\omega)|_{\omega=2\pi k/N}$ when $L \le N$. $X(k) = Nc_k$ (DFS of the periodic extension). Sampling $X(\omega)$ at N points gives $x_p(n) = \sum_m x(n-mN)$: time aliasing iff $N < L$ (write x in rows of N, add columns; T5 Q6). Direct DFT N^2 multiplies; FFT $O(N\log_2 N)$ (radix-2: $\frac{N}{2}\log_2 N$).

DFT properties (all indices mod N)

Property	Time	DFT
Circular shift	$x((n-l))_N$	$X(k)e^{-j2\pi kl/N}$
Circular reversal	$x((-n))_N = x(N-n)$; $\{x_0, x_1, x_2, x_3\} \to \{x_0, x_3, x_2, x_1\}$	$X((-k))_N = X(N-k)$
Frequency shift	$x(n)e^{j2\pi ln/N}$	$X((k-l))_N$
Conjugate	$x^*(n)$	$X^*((-k))_N$. Real x: $X(N-k) = X^*(k)$
Circular convolution	$x_1 \circledast_N x_2 = \sum_{m=0}^{N-1} x_1(m)x_2((n-m))_N$	$X_1(k)X_2(k)$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{N}X_1 \circledast_N X_2$
Circular correlation	$\tilde{r}_{xy}(l) = \sum_n x(n)y^*((n-l))_N$	$X(k)Y^*(k)$
Parseval	$\sum x(n)y^*(n)$	$\frac{1}{N}\sum X(k)Y^*(k)$
Checks	$X(0) = \sum x(n)$; $x(0) = \frac{1}{N}\sum X(k)$; $\sum\|x\|^2 = \frac{1}{N}\sum\|X\|^2$	$X(N/2) = \sum(-1)^n x(n)$ (N even)
Deconvolution	$X(k) = Y(k)/H(k)$	fails where $H(k) = 0$

Circular vs linear convolution (x length L, h length M)

	Linear $x * h$	**Circular** $x \circledast_N h$
Sum	$\sum_m x(m)h(n-m)$	$\sum_{m=0}^{N-1} x(m)h((n-m))_N$
Length	$L+M-1$	exactly N
By hand	multiplication table	circulant matrix (column 0 = h, shift down and wrap), or linear then **fold** samples $n \geq N$ back by N and add
Relation	$y_c(n) = \sum_r y_{lin}(n+rN)$, $0 \leq n \leq N-1$	equal iff $N \geq L+M-1$
N too small	first $L+M-1-N$ samples are corrupted	rest are correct
Long inputs (s25–27)	OLA: blocks of L, pad to $N = L+M-1$, **add** $M-1$ overlaps	OLS: keep last $M-1$ inputs, **discard** first $M-1$ outputs

Block convolution layouts: overlap-add and overlap-save

Resolution and spectral analysis (L06 slides 29–32, T5 Q5)

Quantity	Formula	Note
Sampling	$F_s \geq 2B$, $T_s = 1/F_s$	anti-alias filter first
Record length	$T_0 = L/F_s$ (L real samples)	
Resolution	$\Delta F = 1/T_0 = F_s/L$	ability to separate two tones
Bin spacing	F_s/N; bin k: $F_k = kF_s/N$, $\omega_k = 2\pi k/N$	$k > N/2$ means $-(N-k)F_s/N$
Design (T5 Q5)	$N \geq F_s/\Delta F$, round **up** to 2^m; $T_0 = N/F_s$	course uses F_s/N with $N = L$
Zero padding	finer grid only ("better resolution for sampled DTFT", slide 9)	does **not** separate closer tones
Real tone at F_0	peaks at $k_0 = F_0 N/F_s$ and $N - k_0$; aligned: $\lvert X(k_0)\rvert = AN/2$	non-integer k_0: leakage
Hanning (slide 32)	$w(n) = \frac{1}{2}\left(1 - \cos\frac{2\pi n}{L-1}\right)$, $0 \leq n < L$	less leakage, main lobe about $2\times$ wider

L07 z-transform

$X(z) = \sum x(n) z^{-n}$. Always give the ROC: $X(z)$ alone does not fix $x(n)$.

z-transform pairs

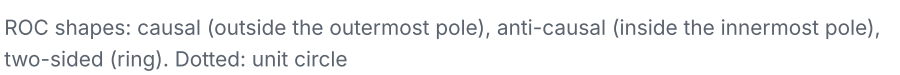
$x(n)$	$X(z)$	ROC
$\delta(n)$	1	all z
$\delta(n-d)$	z^{-d}	all z except 0 ($d > 0$) or ∞ ($d < 0$)
$u(n)$	$\dfrac{1}{1 - z^{-1}}$	$\lvert z\rvert > 1$
$a^n u(n)$	$\dfrac{1}{1 - az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u(-n-1)$	$\dfrac{1}{1 - az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u(n)$	$\dfrac{az^{-1}}{(1 - az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-na^n u(-n-1)$	$\dfrac{az^{-1}}{(1 - az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
$(n+1)a^n u(n)$	$\dfrac{1}{(1 - az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$a^n \cos(\omega_0 n) u(n)$	$\dfrac{1 - a\cos\omega_0\, z^{-1}}{1 - 2a\cos\omega_0\, z^{-1} + a^2 z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$
$a^n \sin(\omega_0 n) u(n)$	$\dfrac{a\sin\omega_0\, z^{-1}}{1 - 2a\cos\omega_0\, z^{-1} + a^2 z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$
$u(n) - u(n-M)$	$\dfrac{1 - z^{-M}}{1 - z^{-1}}$	$z \neq 0$

z-transform properties (L07 slides 10–14)

Property	Time	z-domain	ROC
Linearity	$Ax + By$	$AX + BY$	at least $R_x \cap R_y$
Shift	$x(n-d)$	$z^{-d}X(z)$ ($x(n+4) \to z^{+4}X$)	R_x, ± 0 or ∞
z-scaling	$z_0^n x(n)$; $e^{j\omega_0 n}x(n)$	$X(z/z_0)$; $X(e^{-j\omega_0}z)$	$\lvert z_0\rvert R_x$
Reversal	$x(-n)$	$X(z^{-1})$	$1/R_x$: invert **and** swap radii
Time expansion	$x_{(k)}(n)$	$X(z^k)$	$R_x^{1/k}$
Derivative	$n\,x(n)$	$-z\,dX/dz$	R_x
Conjugate	$x^*(n)$	$X^*(z^*)$	R_x
Convolution	$h * x$	$H(z)X(z)$	at least $R_h \cap R_x$
Initial value	causal x only	$x(0) = \lim_{z\to\infty} X(z)$	

ROC rules (L07 slide 7, slide 17)

Signal	ROC
Any	a ring centred on 0; contains no poles; bounded by poles
Finite length	whole plane, except possibly $z = 0$ (if any $n > 0$ sample) and/or $z = \infty$ (if any $n < 0$ sample)
Causal (right-sided from $n \geq 0$)	$\lvert z\rvert >$ outermost pole (includes ∞)
Anti-causal	$\lvert z\rvert <$ innermost pole
Two-sided	ring between two pole radii, or empty (no transform)
DTFT exists	ROC contains $\lvert z\rvert = 1$; then $X(\omega) = X(e^{j\omega})$, $X(0) = X(1)$, $X(\pi) = X(-1)$



ROC shapes: causal (outside the outermost pole), anti-causal (inside the innermost pole), two-sided (ring). Dotted: unit circle

Poles and zeros: write in positive powers, $X = \frac{b_0}{a_0} z^{N-M} \frac{\prod(z-z_k)}{\prod(z-p_k)}$. $N > M$: $(N-M)$ zeros at 0; $M > N$: poles at 0. Pole at ∞ if $X(\infty) = \infty$, zero at ∞ if $X(\infty) = 0$. Count check: poles = zeros including 0 and ∞. Block diagrams: cascade $H_1 H_2$; parallel $H_1 \pm H_2$; feedback $\frac{G}{1+GF}$ (minus at adder), $\frac{G}{1-GF}$ (plus). Cascade with a non-causal block (T6 Q3, z^{+1}) is non-causal. Any z^{+k} term in the expanded $H(z)$ means not causal.

L08 Inverse z-transform and LTI analysis

Partial-fraction term to sequence (L08 slide 4). Work in $w = z^{-1}$ with monic factors $(1 - p_i z^{-1})$.

Term	ROC outside pole ($\lvert z\rvert > \lvert p\rvert$)	ROC inside pole ($\lvert z\rvert < \lvert p\rvert$)
$\frac{A}{1 - pz^{-1}}$	$Ap^n u(n)$	$-Ap^n u(-n-1)$
$\frac{A}{(1 - pz^{-1})^2}$	$A(n+1)p^n u(n)$	$-A(n+1)p^n u(-n-1)$
complex pair $p = re^{j\theta}$, residues A, A^*	$2\lvert A\rvert r^n \cos(\theta n + \angle A)u(n)$	$-2\lvert A\rvert r^n \cos(\theta n + \angle A)u(-n-1)$
constant K (improper, after division)	$K\delta(n)$	$K\delta(n)$

Residue (cover-up): $A_i = \left[(1 - p_i z^{-1}) X(z)\right]_{z=p_i}$. Equal z^{-1} degrees: K = ratio of the **highest**-power coefficients (num/den). Check: $\sum A_i + K = X(z \to \infty)$. N distinct pole radii give $N + 1$ possible ROCs, so $N + 1$ possible signals (T7 Q1).

Causality and stability from $H(z)$ (L08 slide 7)

Property	ROC condition	Pole condition
Causal	exterior of a circle, including ∞	ROC outside outermost pole
Anti-causal	interior of a circle	ROC inside innermost pole
Stable	ROC contains $\lvert z\rvert = 1$	
Causal and stable		all poles inside $\lvert z\rvert = 1$
Anti-causal and stable		all poles outside $\lvert z\rvert = 1$
FIR (MA, all-zero)	M zeros, Mth-order pole at 0	always stable
All-pole (AR)	N poles, Nth-order zero at 0	

LCCDE $\leftrightarrow H$: $y(n) + \sum_{k=1}^{N} a_k y(n-k) = \sum_{k=0}^{M} b_k x(n-k) \iff H(z) = \dfrac{\sum b_k z^{-k}}{1 + \sum a_k z^{-k}}$. The LCCDE alone does not fix the ROC. System ID: $H = Y/X$, factor and cancel; ROC from "causal" or "stable"; $R_y \supseteq R_h \cap R_x$ (T7 Q3). Power series: the coefficient of z^{-n} is $x(n)$; causal ROC: long-divide in ascending powers of z^{-1}; anti-causal: ascending powers of z.

Unilateral z-transform (L08 slides 13–14). $X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$. The inverse is always causal (no ROC choice).

Time	$\mathcal{Z}^+$
$y(n-1)$	$z^{-1}Y^+(z) + y(-1)$
$y(n-2)$	$z^{-2}Y^+(z) + z^{-1}y(-1) + y(-2)$
$y(n-N)$	$z^{-N}Y^+(z) + z^{-(N-1)}y(-1) + \cdots + y(-N)$
$x(n+1)$	$zX^+(z) - zx(0)$
$x(n+N)$	$z^N X^+(z) - z^N x(0) - z^{N-1}x(1) - \cdots - zx(N-1)$
causal input $x(n-k)$	$z^{-k}X(z)$ (no IC terms, since $x(-1) = x(-2) = 0$)

Second order, $y(n) + a_1 y(n-1) + a_2 y(n-2) = \ldots$: $A(z)Y + I(z) = B(z)X$ with $I(z) = a_1 y(-1) + a_2 y(-2) + a_2 y(-1) z^{-1}$. Then $Y_{zs} = BX/A$ (input and system poles), $Y_{zi} = -I/A$ (system poles only).

L09 Digital filters and pole-zero patterns

$\omega_0 = 2\pi F_0 / F_s$; back to Hz: $F = \omega F_s / 2\pi$; rad/s: $\Omega = \omega F_s$.

Filter	$H(z)$	Pole-zero pattern	Key facts
Lowpass		poles near $z = 1$; zeros near/on circle near $z = -1$	
Highpass	$H_{HP}(z) = H_{LP}(-z)$; $h_{HP}(n) = (-1)^n h_{LP}(n)$	poles near $z = -1$; zeros near $z = 1$	flip sign of odd powers of z^{-1}, num **and** den
Bandpass		conjugate poles $re^{\pm j\omega_0}$ in the passband	
Resonator (s15)	$\frac{b_0(1 - z^{-2})}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	poles $re^{\pm j\omega_0}$, zeros ± 1 (or both at 0)	peak near ω_0; $r \to 1$ sharper
FIR notch (s17)	$b_0(1 - 2\cos\omega_0\, z^{-1} + z^{-2})$	zeros $e^{\pm j\omega_0}$, double pole at 0	wide notch; normalise b_0 at the stated ω
Pole-zero notch (extension, not on slides)	$b_0 \frac{1 - 2\cos\omega_0\, z^{-1} + z^{-2}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	zeros $e^{\pm j\omega_0}$, poles $re^{\pm j\omega_0}$	-3 dB width $\approx 2(1 - r)$ rad (rule of thumb)
Comb / moving average (s19)	$\frac{1}{M+1}\sum_{k=0}^{M} z^{-k} = \frac{1}{M+1}\frac{1 - z^{-(M+1)}}{1 - z^{-1}}$	M zeros $e^{j2\pi k/(M+1)}$, $k = 1..M$; M poles at 0	nulls $F_k = kF_s/(M+1)$; $\lvert H\rvert = \frac{1}{M+1}\left\lvert\frac{\sin((M+1)\omega/2)}{\sin(\omega/2)}\right\rvert$, phase $-M\omega/2$
Allpass (s21)	$z^{-N}\frac{A(z^{-1})}{A(z)}$, $A(z) = \sum_{k=0}^{N} a_k z^{-k}$, $a_0 = 1$	poles and zeros are reciprocals	$\lvert H\rvert = 1$; phase equaliser
1st-order allpass	$\frac{a + z^{-1}}{1 + az^{-1}}$	pole $-a$, zero $-1/a$	$\tau(\omega) = \frac{1 - a^2}{1 + 2a\cos\omega + a^2}$
Oscillator (s23)	$\frac{b_0}{1 - 2\cos\omega_0\, z^{-1} + z^{-2}}$	poles $e^{\pm j\omega_0}$ on the circle	$h(n) = \frac{b_0}{\sin\omega_0}\sin(\omega_0(n+1))u(n)$ ($b_0 = \sin\omega_0$ gives $\sin(\omega_0(n+1))$); marginally unstable
2-tap/3-tap FIR by hand	$b_0 + b_1 z^{-1} + b_2 z^{-2}$	set $H(\omega_0) = 0$ for a null	DC gain 1: $\sum b = 1$; gain 1 at π: $b_0 - b_1 + b_2 = 1$; linear phase: $b_0 = b_2$

Phase, delay and inverse systems (L09 s8–9, T8 Q4–Q5)

Item	Rule
Linear phase	$\theta(\omega) = -\omega n_0$; group delay $\tau(\omega) = -d\theta/d\omega = n_0$; $\pm\pi$ jumps where the real amplitude changes sign are still linear phase
Inverse	$H_I = 1/H$: zeros of H become poles of H_I
Minimum phase	all zeros inside $\lvert z\rvert = 1$: stable inverse is **causal** (T8 Q4)
Maximum phase	all zeros outside: stable inverse is **anti-causal**
Nonminimum phase (mixed)	stable inverse is **non-causal**, ROC is the ring containing $\lvert z\rvert = 1$ (T8 Q5)
Zero on the circle	no stable inverse (notch, comb)

L10 Filter design (IIR and FIR)

IIR: bilinear transform, Butterworth, Chebyshev

Item	Formula	Note
Bilinear map	$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}, z = \frac{1+sT/2}{1-sT/2}$	LHP maps inside circle: stable, no aliasing; take $T=1$ (it cancels)
Warping	$\Omega = \frac{2}{T}\tan\frac{\omega}{2}, \omega = 2\arctan\frac{\Omega T}{2}$	$s=0\to z=1$, $s=\infty\to z=-1$
Prewarp	$\Omega_i = \frac{2}{T}\tan\frac{\omega_i}{2}, \omega_i = 2\pi F_i/F_s$	do this **before** any ratio Ω_s/Ω_p
dB to linear	$\delta_p = 10^{-A_p/20}, \delta_s = 10^{-A_s/20}$	1 dB: $\delta_p = 0.8913$; 20 dB: $\delta_s = 0.1$
Factor method, $K = 2/T$	$s + a \to \frac{(K+a)+(a-K)z^{-1}}{1+z^{-1}}$	numerator $\Omega_c^N(1+z^{-1})^N$ for all-pole $H(s)$
	$s^2 + bs + c \to \frac{(K^2+bK+c)+2(c-K^2)z^{-1}+(K^2-bK+c)z^{-2}}{(1+z^{-1})^2}$	divide by the constant so $a_0 = 1$
Impulse invariance (low priority)	$h(n) = Th_a(nT)$; pole $p \to e^{pT}$	aliases; bilinear avoids this

Butterworth (L10 slides 8–18)

Item	Formula
Response	$\lvert H(\Omega)\rvert^2 = \frac{1}{1+(\Omega/\Omega_c)^{2N}}$; Ω_c = 3-dB point = pole radius; $-20N$ dB/dec
Order (round **up**)	$N = \frac{\log\left[\frac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}\right]}{2\log(\Omega_s/\Omega_p)} = \frac{\log\left[(10^{A_s/10}-1)/(10^{A_p/10}-1)\right]}{2\log(\Omega_s/\Omega_p)}$
Ω_c, passband exact (slide 17)	$\Omega_c = \Omega_p\left(\frac{\delta_p^2}{1-\delta_p^2}\right)^{1/2N} = \frac{\Omega_p}{(10^{A_p/10}-1)^{1/2N}}$
Ω_c, stopband exact (T9 Q5)	$\Omega_c = \Omega_s\left(\frac{\delta_s^2}{1-\delta_s^2}\right)^{1/2N} = \frac{\Omega_s}{(10^{A_s/10}-1)^{1/2N}}$ (say which one you used)
Poles	$s_k = \Omega_c e^{j(\pi/2+\pi(2k+1)/2N)}, k = 0..N-1$; $H(s) = \frac{\Omega_c^N}{\prod(s-s_k)}$; $H(0) = 1$
Denominators	$N=2$: $s^2 + 1.4142\Omega_c s + \Omega_c^2$. $N=3$: $(s+\Omega_c)(s^2+\Omega_c s+\Omega_c^2)$. $N=4$: $(s^2+0.7654\Omega_c s+\Omega_c^2)(s^2+1.8478\Omega_c s+\Omega_c^2)$. Any conjugate pair at angle θ_k: $s^2 - 2\Omega_c\cos\theta_k\, s + \Omega_c^2$
Order from pole radius (T9 Q6)	$\Omega_c = R$; $N = \frac{\log(10^{A/10}-1)}{2\log(\Omega_1/\Omega_c)}, \Omega_1 = 2\pi F_1$ (pure analogue)

Chebyshev (L10 slides 20–27). The slides print "$\cos^{-1}$" in the order formula. Both arguments exceed 1, so use $\cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right)$.

Item	Formula
Type I response	$\lvert H\rvert^2 = \dfrac{1}{1+\epsilon^2 T_N^2(\Omega/\Omega_p)}$; equiripple passband, monotonic stopband, all-pole; $T_N(x) = \cos(N\cos^{-1}x)$, $\lvert x\rvert \le 1$
Type II response	$\lvert H\rvert^2 = \dfrac{1}{1+\epsilon^2\left[T_N^2(\Omega_s/\Omega_p)/T_N^2(\Omega_s/\Omega)\right]}$; monotonic passband, equiripple stopband; poles and zeros (zeros at $\Omega = \Omega_s/\cos\frac{(2k+1)\pi}{2N}$)
ϵ (both types, from passband)	$\epsilon = \sqrt{1/\delta_p^2 - 1} = \sqrt{10^{A_p/10}-1}$. 1 dB: 0.5088; 0.5 dB: 0.3493
Order (both types, round **up**)	$N = \dfrac{\cosh^{-1}\left(\dfrac{\sqrt{1-\delta_s^2}}{\epsilon\,\delta_s}\right)}{\cosh^{-1}(\Omega_s/\Omega_p)}$
DC gain, Type I	1 if N odd; $\delta_p = 1/\sqrt{1+\epsilon^2}$ if N even
Type I poles (slide 22)	$\beta = \left(\dfrac{\sqrt{1+\epsilon^2}+1}{\epsilon}\right)^{1/N}$; $r_1 = \Omega_p\dfrac{\beta^2+1}{2\beta}$, $r_2 = \Omega_p\dfrac{\beta^2-1}{2\beta}$; $\phi_k = \dfrac{\pi}{2} + \dfrac{(2k+1)\pi}{2N}$; $p_k = r_2\cos\phi_k + jr_1\sin\phi_k$; $H(s) = K/\prod(s-p_k)$, $K = \prod(-p_k)$ (N odd), $\prod(-p_k)/\sqrt{1+\epsilon^2}$ (N even)
Family ranking	order needed: Elliptic < Chebyshev I = II < Butterworth < Bessel. Phase: Bessel best (linear in passband, lost after mapping) ... Elliptic worst

Frequency transformations (L10 slides 47–48, low priority). Analogue, from a prototype LPF with edge Ω_p: LP $s \to \frac{\Omega_p}{\Omega_p'}s$; HP $s \to \frac{\Omega_p\Omega_p'}{s}$; BP $s \to \Omega_p\frac{s^2+\Omega_l\Omega_u}{s(\Omega_u-\Omega_l)}$ (order doubles); BS $s \to \Omega_p\frac{s(\Omega_u-\Omega_l)}{s^2+\Omega_l\Omega_u}$. Digital LP to HP: $z^{-1} \to -\frac{z^{-1}+a}{1+az^{-1}}$, $a = -\frac{\cos((\omega_p+\omega_p')/2)}{\cos((\omega_p-\omega_p')/2)}$. Other digital maps: slide 48.

FIR: windows and linear phase

Windows as the course defines them (L10 slide 41), all for $0 \le n \le M-1$, zero otherwise. Peak sidelobe, main-lobe width and filter stopband figures are standard textbook values (slides 42–44 show them as plots); the stopband figures were checked numerically.

Window	$w(n)$	$M=5$ values	Peak sidelobe	Main lobe	FIR stopband
Rectangular	1	$\{1,1,1,1,1\}$	-13 dB	$4\pi/M$	-21 dB
Bartlett	$1 - \dfrac{2\lvert n-(M-1)/2\rvert}{M-1}$	$\{0,0.5,1,0.5,0\}$	-27 dB	$8\pi/M$	-26 dB
Hanning	$\frac{1}{2}\left(1-\cos\dfrac{2\pi n}{M-1}\right)$	$\{0,0.5,1,0.5,0\}$	-32 dB	$8\pi/M$	-44 dB
Hamming	$0.54 - 0.46\cos\dfrac{2\pi n}{M-1}$	$\{0.08,0.54,1,0.54,0.08\}$	-43 dB	$8\pi/M$	-53 dB
Blackman	$0.42 - 0.5\cos\dfrac{2\pi n}{M-1} + 0.08\cos\dfrac{4\pi n}{M-1}$	$\{0,0.34,1,0.34,0\}$	-58 dB	$12\pi/M$	-75 dB

Choose the lightest window whose stopband figure meets A_s. Wider main lobe means a wider transition band (needs larger M).

Ideal impulse responses (shift by α before windowing): LPF $h_d(n) = \frac{\sin \omega_c n}{\pi n}$ ($h_d(0) = \omega_c/\pi$; course form $\frac{\omega_c}{\pi}\text{sinc}(\omega_c n)$, numpy `(wc/pi)*np.sinc(wc*n/pi)`). HPF $\delta(n) - \frac{\sin \omega_c n}{\pi n}$. BPF $\frac{\sin \omega_2 n - \sin \omega_1 n}{\pi n}$.

Linear-phase FIR types (L10 slides 33–36; type numbers are the textbook names).
$\alpha = (M-1)/2$ = group delay for all four.

Type	Symmetry	M	$H(\omega)$	Forced zeros	Cannot make
I	$h(n) = h(M-1-n)$	odd	$e^{-j\alpha\omega}\left[h(\alpha) + 2\sum_{n=0}^{\alpha-1} h(n)\cos((\alpha-n)\omega)\right]$	none	(any type OK)
II	symmetric	even	$e^{-j\alpha\omega}\, 2\sum_{n=0}^{M/2-1} h(n)\cos((\alpha-n)\omega)$	$z=-1$ ($H(\pi)=0$)	HP, BS
III	$h(n) = -h(M-1-n)$	odd	$je^{-j\alpha\omega}\, 2\sum_{n=0}^{\alpha-1} h(n)\sin((\alpha-n)\omega)$, $h(\alpha)=0$	$z=\pm 1$	LP, HP, BS
IV	antisymmetric	even	$je^{-j\alpha\omega}\, 2\sum_{n=0}^{M/2-1} h(n)\sin((\alpha-n)\omega)$	$z=1$ ($H(0)=0$)	LP, BS

Antisymmetric phase is $-\alpha\omega + \pi/2$ (plus $\pm\pi$ jumps). Filling a missing sample (T9 Q8): compare end samples to decide symmetric or antisymmetric, then mirror about α.

Procedures

P1–P15 cover every Sample Test 1 question type (L02–L04). P16–P22 cover L05–L10.

P1. Sample a CT signal; find T_s and samples (ST1 Q1a, T1 Q2)

1. $T_s = 1/F_s$. List every F_0: divide the t coefficient by 2π ($\sin(480\pi t)$ is 240 Hz). Nyquist rate $= 2F_{max}$.
2. Substitute $t = nT_s$: each term becomes $\omega_0 = 2\pi F_0/F_s$.
3. Reduce each ω_0 into $(-\pi, \pi]$; use $\cos(2\pi k n) = 1$, $\cos(\pi n) = (-1)^n$.
4. Tidy negative frequencies ($\sin$ flips sign) and combine like terms.
5. Evaluate the requested samples after reducing. Say whether aliasing occurred (F_0 vs $F_s/2$).
6. Ideal DAC, if asked: $y(t) = x(n)|_{n=F_s t}$ with the reduced ω.

Check: ST1 Q1(a) $T_s = 0.01$ s, $x(n) = 10e^{-0.0001n}$: $10, 9.9990, 9.9980, 9.9970$. T1 Q2: $x(n) = -2\sin(0.8\pi n)$, $y(t) = -2\sin(480\pi t)$.

P2. Periodic? Fundamental period (T1 Q1)

1. CT: always periodic, $T_0 = 2\pi/|\Omega_0|$.
2. DT: for each component $f_i = \omega_i/2\pi$. Any irrational f_i (leftover π): aperiodic, stop.
3. Write $f_i = k_i/N_i$ in lowest terms; $N = \text{LCM}(N_i)$. Products: same rule (both factors must be periodic).

Check: T1 Q1(e) $f = \frac{1}{4}, \frac{1}{16}, \frac{1}{8}$: $N = 16$. Reference self-test: $\cos(3\pi n/7) + \sin(\pi n/4)$: $N = \text{LCM}(14, 8) = 56$.

P3. ADC sizing (T1 Q3–Q4)

1. Range $= x_{max} - x_{min}$ ($2A$ for $A\cos$).
2. Given Δ: $L = \text{range}/\Delta + 1$, $b = \lceil \log_2 L \rceil$. Given b: $L = 2^b$, $\Delta = \text{range}/(2^b - 1)$.
3. Bit rate bF_s; max frequency $F_s/2$; max error $\Delta/2$.

P4. Stem plot from steps and impulses (ST1 Q1b)

1. Turn each step difference into a window: $u(n+a) - u(n-b)$ is 1 on $-a \le n \le b-1$.
2. Evaluate the envelope only on the window. Add each $\delta(n-d)$ as a spike at $n = d$.
3. Tabulate n against $x(n)$, including zeros inside the range. Plot with labelled axis and values.

Check: ST1 Q1(b) $x(n) = \{2, \underset{\uparrow}{1}, 0.5, 0, 1\}$, $n = -1..3$ (note $x(2) = 0$).

P5. Sequence operations and energy (ST1 Q2)

1. Write the index under every sample (ST1: $x(0) = 1$, the underlined third sample, so $n = -2..3$).
2. Shifts: add the delay to every index; for sums, one row per copy, add columns.
3. Upsample $x(n/2)$: index $m \to 2m$, zeros between. Downsample $x(2n)$: keep even indices, halve them.
4. $(-1)^n$: negate odd-index samples (check the index, not the position in the list).
5. $x(\alpha n + \beta)$: substitute each n directly.
6. Energy $= \sum x^2$; finite and nonzero means an energy signal.

P6. System properties, tick boxes (ST1 Q3, T2 Q1, T3 Q1)

1. **Static?** Any index other than n, any sum, any $y(n-k)$: dynamic.
2. **Linear?** $x = 0 \Rightarrow y = 0$? A constant term or x^2, $|x|$, $\cos x$ fails. n-dependent gains are fine.
3. **TI?** Two columns: (i) replace $x(\cdot)$ by $x(\cdot - d)$, leave explicit n; (ii) replace every n by $n - d$. Compare. Explicit n coefficient, $u[n]$ factor, $x(2n)$, $x(-n)$: time-varying.
4. **Causal?** Any $x(n+k)$, $k > 0$, at any n (check negative n for $x(-n)$).
5. **Stable?** Bound $|y|$ by M_x, or use $x = u(n)$ to blow it up.
6. Write a two-word reason beside every tick.

P7. Linearity from observed input-output pairs (T2 Q2)

1. Use time invariance to shift one known pair so its input lines up with another input.
2. Test whether the inputs are scalar multiples or sums of each other.
3. If linear, the outputs must be related the same way. One mismatch proves nonlinear. Matching pairs prove nothing in general.

P8. Correlation and delay (L03 s15–18, T3 Q7)

1. State the definition. Lecture: $r_{xy}(l) = \sum x(n)y(n-l)$. Tutorial: $R_{xy}(l) = \sum x(k)y(k+l) = x(-n) * y(n)$.
2. Reverse the first signal (indices $[n_1, n_2] \to [-n_2, -n_1]$) and long-multiply with the second.
3. Label the lag axis: start lag = $-n_2 + m_1$ (tutorial form).
4. Check $r_{xx}(0) = E_x$, r_{xx} even. Delay $\hat{d} = \arg\max |r_{yx}(l)|$; range $r = c\hat{d}/(2F_s)$.

P9. Characteristic modes (ST1 Q4a)

1. Copy the y coefficients into $\lambda^k + a_1\lambda^{k-1} + \cdots + a_k = 0$.
2. Solve; for complex roots give r and θ.
3. Write the modes by root type (L04 table). No $y(n-k)$ terms: no modes.
4. One line on stability: all $|\lambda| < 1$ stable; $|\lambda| = 1$ or more is unstable.

Check: ST1 Q4(a) $\lambda = \frac{1 \pm j\sqrt{3}}{2} = e^{\pm j\pi/3}$; modes $\cos(\pi n/3)$, $\sin(\pi n/3)$; $|\lambda| = 1$, not BIBO stable.

P10. Output by recursion (ST1 Q4b)

1. Rearrange: $y(n) = -\sum a_k y(n-k) + \sum b_k x(n-k)$ (flip the y signs).
2. State the ICs ($y(n) = 0$ for $n < 0$, or the given values).
3. Table: n | $x(n)$ | arithmetic | $y(n)$. Give 4–5 samples; mark $n = 0$.

Check: ST1 Q4(b) $y = \{\underset{\uparrow}{1}, 5.25, 5.3125, 0.328125, -4.917969, \ldots\}$.

P11. Full solution $y_h + y_p$ with ICs (T3 Q3, Q4)

1. Modes, then $y_h = \sum C_p \lambda_p^n$ (with $n\lambda^n$ for repeats).
2. Trial y_p of the input's form; substitute into the **full** LCCDE; cancel the common factor (e.g. 3^{-n}, using $3^{-(n-1)} = 3 \cdot 3^{-n}$); solve for K.
3. $y(0), \ldots, y(k-1)$ by recursion from the ICs.
4. Solve $y_h + y_p$ at $n = 0..k-1$ for the C_p.
5. Attach $u(n)$ or "$n \geq 0$"; check $y(k)$ against the recursion.

Check: T3 Q3 $K = 16$, $y(0) = 1$, $y(1) = \frac{1}{12}$, $C_1 = -6$, $C_2 = -9$. T3 Q4 $K = 0.25$, $y(0) = 1.5$, $y(1) = -0.75$, $C_1 = 1.25$, $C_2 = 0.5$.

P12. Impulse response from an LCCDE (ST1 Q4c, T3 Q5)

1. Set $x = \delta(n)$, all ICs zero.

2. No feedback: $h(n) = \{b_0, \ldots, b_M\}$, done.
3. Otherwise $h(n) = \sum C_p \lambda_p^n$ (complex pair: $A\cos\theta n + B\sin\theta n$, times r^n).
4. Recursion for $h(0), \ldots, h(k-1)$; solve for the constants. If $m \geq k$ add δ terms.
5. Check one more sample; classify causal / stable / FIR or IIR.

Check: ST1 Q4(c) $h(0) = 1$, $h(1) = 5$,
$h(n) = [\cos\frac{\pi n}{3} + 3\sqrt{3}\sin\frac{\pi n}{3}]u(n) = \{1, 5, 4, -1, -5, -4, \ldots\}$ (period 6), causal, IIR, unstable.

P13. Finite convolution, tabular (T3 Q6a)

1. Output start $= n_x + n_h$; length $N_x + N_h - 1$.
2. Long multiplication: rows $x(k)h$ shifted k places; add columns.
3. Check $\sum y = (\sum x)(\sum h)$; mark $n = 0$.

Check: T3 Q6(a) $\{2, -3, -3, -5, 6, 5, 10, 12\}$; $\sum = 24 = 3 \times 8$.

P14. Closed-form convolution (ST1 Q4d, T2 Q3, T3 Q6b–d)

1. Write $y(n) = \sum_k h(k)x(n-k)$ with the steps kept in.
2. Steps to limits: $u(k)$, $u(n-k)$ give $0 \leq k \leq n$ and a $u(n)$ on the answer. Finite windows: one case per overlap stage.
3. Pull out k-free factors; write as $\sum r^k$; geometric box.
4. Simplify; check $y(0)$ and $y(1)$ directly.

Check: ST1 Q4(d) (with the given $h = (2^{-n} + 3^{-n})u(n)$, not the answer to (c)):
$y = (2 \cdot 2^{-n} + 4 \cdot 3^{-n} - 4 \cdot 4^{-n})u(n)$; $y(0) = 2$, $y(1) = \frac{4}{3}$. T3 Q6(b):
$3^n \frac{1-6^{-(n+1)}}{1-6^{-1}} = (1.2 \cdot 3^n - 0.2 \cdot 2^{-n})u(n)$.

P15. Interconnected LTI systems (T2 Q4)

1. Inside out: series blocks convolve, branches that add (or subtract) add (or subtract).
2. Use $x * \delta(n-d) = x(n-d)$; simplify step differences sample by sample.
3. Classify the overall h: causal if $h(n) = 0$ for $n < 0$; stable if $\sum|h| < \infty$.

Check: T2 Q4 $h = 2u(n) + 3u(n-1) + 3u(n-2) + 2u(n-3) = \{2, 5, 8, 10, 10, \ldots\}$: causal, unstable.

P16. Steady-state response to sinusoids (L05 s29–30, T4 Q3–Q5)

1. Get $H(\omega)$: from $h(n)$, from the LCCDE (all y terms on the left), or $H(z)$ at $z = e^{j\omega}$. Confirm the system is stable; otherwise there is no steady state.
2. List every frequency in $x(n)$. A constant is $\omega = 0$, $(-1)^n$ is $\omega = \pi$. Convert Hz with $\omega = 2\pi F/F_s$ and reduce into $(-\pi, \pi]$.
3. Evaluate $H(\omega_i)$ at each one as a complex number. Write $|H|$ and θ. A negative real value has magnitude $|\cdot|$ and phase $\pm\pi$.
4. Each term: $A\cos(\omega_i n + \phi) \to A|H(\omega_i)|\cos(\omega_i n + \phi + \theta(\omega_i))$.
5. Add the terms. Check $H(0) = \sum h$ and $H(\pi) = \sum(-1)^n h$. For x switched on at $n = 0$, this is the response after the transients decay.

P17. Linear convolution with the DFT (L06 s23–24)

1. Note lengths L (input) and M (filter). Output length is $L + M - 1$.
2. Choose $N \geq L + M - 1$ (next power of 2 if an FFT is used).
3. Zero-pad both sequences to N.
4. Compute $X(k)$ and $H(k)$; form $Y(k) = X(k)H(k)$.
5. IDFT; keep $y(0), \ldots, y(L + M - 2)$.
6. Check $\sum y = (\sum x)(\sum h)$ and $y(0) = x(0)h(0)$. If N was given and is too small, fold the linear result mod N: the first $L + M - 1 - N$ samples are wrong.

P18. Inverse z-transform with ROC (L08 s4, T7)

1. Write X in powers of z^{-1}. If the numerator degree is $\geq$ the denominator degree, divide first (equal degrees give the constant K).
2. Factor the denominator into monic factors $(1 - p_i z^{-1})$; pull out any leading constant.
3. Residues by cover-up: $A_i = [(1 - p_i z^{-1})X]_{z=p_i}$. Double pole: get the squared-term coefficient by cover-up, the other from $z^{-1} = 0$. Check $\sum A_i + K = X(\infty)$.

4. Fix the ROC: given, or "causal" (outside all poles), "stable"/"absolutely summable" (ring containing $|z| = 1$).
5. For each pole: inside the ROC's inner edge gives $A_i p_i^n u(n)$; outside its outer edge gives $-A_i p_i^n u(-n-1)$.
6. Combine complex pairs into $2|A|r^n \cos(\theta n + \angle A)$. State $x(n)$ with the ROC, and causal/stable if asked.

P19. LCCDE with initial conditions by unilateral z (L08 s13–14)

1. Write the LCCDE with all y terms on the left.
2. Take $\mathcal{Z}^+$ of each term with the shift table. Only y terms get IC terms; a causal input has $x(-1) = 0$.
3. Collect: $A(z)Y(z) + I(z) = B(z)X(z)$, so $Y = \frac{B(z)X(z)}{A(z)} - \frac{I(z)}{A(z)}$.
4. Put Y over one denominator in z^{-1}; partial fractions as in P18.
5. Invert every term as causal ($p^n u(n)$).
6. Check $y(0)$ and $y(1)$ against the recursion. Split into y_{zi} (from $-I/A$) and y_{zs} (from BX/A) if asked.

Course check (L08 s14): $y(n) + y(n-1) - 6y(n-2) = 8u(n)$, $y(-1) = y(-2) = -1$ gives $I(z) = 5 + 6z^{-1}$ and $y(n) = (3.2(2^n) + 1.8(-3)^n - 2)u(n)$.

P20. Notch filter from a frequency in Hz (L09 s17, T8 Q3)

1. $\omega_0 = 2\pi F_0 / F_s$ (T8 Q3: $2\pi \cdot 50/500 = \pi/5$).
2. Zeros at $e^{\pm j\omega_0}$: $H(z) = b_0(1 - 2\cos\omega_0\, z^{-1} + z^{-2})$.
3. Normalise at the frequency the question names. DC ($z = 1$): $b_0 = \frac{1}{2 - 2\cos\omega_0}$. $\omega = \pi$ ($z = -1$): $b_0 = \frac{1}{2 + 2\cos\omega_0}$ (T8 Q3: 0.2764).
4. For a narrow notch (if allowed), add poles $re^{\pm j\omega_0}$: denominator $1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}$. Width $\approx 2(1-r)$ rad $= (1-r)F_s/\pi$ Hz. Recompute b_0 at the normalising frequency.
5. Difference equation: $y(n) = b_0[x(n) - 2\cos\omega_0\, x(n-1) + x(n-2)]$ (plus $2r\cos\omega_0\, y(n-1) - r^2 y(n-2)$ for the pole version).
6. Check $H(e^{j\omega_0}) = 0$ and the gain at the normalising frequency.

P21. IIR design, 6 steps (L10 slide 7, T9 Q5)

1. Digital specs: $\omega_p = 2\pi F_{pass}/F_s$, $\omega_s = 2\pi F_{stop}/F_s$; $\delta_p = 10^{-A_p/20}$, $\delta_s = 10^{-A_s/20}$.
2. Prewarp with $T = 1$: $\Omega_p = 2\tan(\omega_p/2)$, $\Omega_s = 2\tan(\omega_s/2)$.
3. Analogue response: Butterworth N (log formula) or Chebyshev ϵ and N ($\cosh^{-1}$ formula). Round N up. Butterworth: choose Ω_c (passband or stopband exact) and say which.
4. $H(s)$: Butterworth poles s_k or the standard denominators; numerator Ω_c^N.
5. Bilinear: substitute $s = 2\frac{1-z^{-1}}{1+z^{-1}}$ (factor method). Normalise $a_0 = 1$ and write $y(n) = -\sum a_k y(n-k) + \sum b_k x(n-k)$.
6. Check the specs: gain in dB $= -10\log_{10}[1 + (\Omega/\Omega_c)^{2N}]$ at Ω_p and Ω_s (equals the digital gain at ω_p, ω_s).

P22. Windowed FIR design (L10 slide 38, T9 Q9)

1. Fix M and $\alpha = (M-1)/2$. Cutoff ω_c given, or the midpoint of ω_p and ω_s (usual choice).
2. Ideal $h_d(n)$ by inverse DTFT (LPF: $\frac{\sin\omega_c n}{\pi n}$, $h_d(0) = \omega_c/\pi$).
3. Choose the window from the stopband attenuation needed.
4. $h(n) = h_d(n-\alpha)w(n)$ for $n = 0, \ldots, M-1$. Evaluate the window with $M-1$ in the cosine.
5. Check symmetry (linear phase) and write $H(\omega) = e^{-j\alpha\omega}A(\omega)$ with $A(\omega) = h(\alpha) + 2\sum_{n<\alpha} h(n)\cos((\alpha - n)\omega)$ for odd M.
6. Check $A(0) = \sum h(n)$ and $A(\omega_c)$, $A(\pi)$ against the spec.

Which method when

Question says	Use	Where
“Advantages of DSP” / “what is a digital signal”	four advantages; DT and discrete-valued	L01
CT signal plus F_s: “find T_s and the samples”	substitute $t = nT_s$, reduce ω	P1
“Nyquist rate”, “what frequencies in $x(n)$”, “reconstructed $y(t)$”	$2F_{max}$; alias F_a; ideal DAC $n = F_s t$	P1, L02
“Is $x(n)$ periodic? Fundamental period?”	rational f_0, LCM	P2
Δ, levels, bits, bit rate	ADC formulas	P3
“Sketch the stem plot” of steps and impulses	windows, then spikes	P4
“Write in sequential form” $x(n-3)+x(n+1)$, $x(n/2)$, $(-1)^n x(n)$	index table	P5
“Energy of $x(n)$” / “energy or power signal”	$\sum\lvert x\rvert^2$; P limit	L02
Even and odd parts	x_e, x_o formulas	L02
“Linear, time-invariant, causal, static, stable?”	five tests with counterexamples	P6
Input-output pairs given, “conclusions about linearity”	shift by TI, test superposition	P7
“Cross-correlation”, “estimate the delay or range”	reverse and convolve; $\arg\max$	P8
“Characteristic modes”	y coefficients only	P9
“Using recursion, evaluate $y(n)$”	recursive form table	P10
“Find the output” of an LCCDE, formula input, ICs given	$y_h + y_p$, constants from recursion	P11
“Find $h(n)$” from an LCCDE	$x = \delta$, zero ICs, modes	P12
Convolve two short sequences	long multiplication	P13
Convolve exponentials or windows in closed form	step limits + geometric sum	P14
Block diagram of LTI systems: “overall $h(n)$”, “causal? stable?”	cascade $*$, parallel $+$	P15
“Find $X(\omega)$” for a finite or one-sided sequence	definition term by term, or geometric series after shifting the index to start at 0	L05 pairs
“Without computing $X(\omega)$, find $X(0)$, $\int X$, $\int\lvert X\rvert^2$”	shortcuts	L05
Output for a sinusoid or sum of sinusoids, stable system	$H(\omega)$ at each frequency	P16
Sketch $\lvert H\rvert$ / decide LP, HP, BP from a plot	pole-zero geometry	L05, L09

Question says	Use	Where
Small DFT ($N \leq 4$)	$\mathbf{W}_N$ matrix	L06
"Circular convolution"	circulant matrix, or linear then fold	L06
Linear convolution "using the DFT"	zero-pad to $N \geq L + M - 1$	P17
Very long input through an FIR	overlap-add or overlap-save	L06
Choose F_s, N, T_0 for a resolution	resolution table	L06
"Find $X(z)$ and ROC"	pairs + properties; ROC rules	L07
"All possible signals" / "absolutely summable" / "causal"	partial fractions + ROC choice	P18
Transfer function from input-output pair	$H = Y/X$	L08
Causal? Stable? from $H(z)$	ROC and pole table	L08
LCCDE with nonzero $y(-1)$, $y(-2)$, z-domain asked	unilateral z	P19
LCCDE at rest, closed-form $y(n)$	bilateral z (P18) or L04 time domain (P11)	P18, P11
Remove a hum at F_0 Hz	notch	P20
Remove all harmonics / nulls at multiples	comb, $F_k = kF_s/(M+1)$	L09
Change phase only / equalise delay	allpass	L09
Stable inverse of a channel	min/max/nonmin phase, then P18	L09
"Convert $H(s)$ to $H(z)$"	bilinear, factor method	L10
Digital IIR from dB specs	6 steps, Butterworth or Chebyshev	P21
"Find ϵ and N"	Chebyshev table ($\cosh^{-1}$)	L10
Linear-phase FIR of length M from ideal response	window method	P22
Make $h(n)$ linear phase / which filters possible	FIR type table	L10

Traps

L01

> **"Digital" means both.** A sampled but unquantised signal is discrete-time, not digital.

L02

> **Hz from the argument.** $\sin(480\pi t)$ is 240 Hz: divide by 2π.

> **Reduce before you use ω.** Reconstructing with 1.2π unreduced just returns $x(t)$; evaluating $\cos(4\pi n)$ with rounding instead of writing 1 (ST1 Q1a).

> **Aliased sine flips sign.** $3\sin(1.2\pi n) = -3\sin(0.8\pi n)$ (T1 Q2).

> **Rate vs frequency.** Nyquist **rate** $= 2F_{max}$; folding frequency $= F_s/2$.

> $\cos(5n)$ **is aperiodic**, $\cos(5t)$ is periodic. One aperiodic component makes a sum aperiodic.

Δ **uses** $L-1$, and b rounds **up** (636 levels: 10 bits).

Missing arrow. Every sequence answer marks $n=0$. ST1 Q2: $x(0)=1$ (underlined), so $(-1)^n$ negates $n=-1,1,3$.

Wrong order in $x(\alpha n+\beta)$**.** Shift by β first, then scale; or substitute n directly. Upsampled $x(n/2)$ keeps every sample and adds zeros; $x(2n)$ loses samples.

L03

$\cos(0.2\pi)$ **is a constant**, so $\cos(0.2\pi)x[n]$ is time-invariant. $\cos(0.2\pi n)x[n]$ is not.

"Looks linear". $y=1+2x$ and $x[n+1]-1$ fail $x=0\Rightarrow y=0$; $nx(n)$, $x(2n)$ are linear (only time-varying).

Causality at negative n**.** $x(-n)$ and $x(-n+2)$ use future inputs for $n<0$ (or $n\le 0$).

Correlation convention. Lecture $r_{xy}(l)$ and tutorial $R_{xy}(l)$ are lag-reversed; T3 Q7's list read with the lecture definition is backwards. State yours.

L04

Modes from the x **side.** Only y coefficients enter; no feedback means no modes (say so).

Recursion sign flip. $y(n)-y(n-1)+y(n-2)$ becomes $y(n)=y(n-1)-y(n-2)+\dots$

ICs into the closed form. Fit constants at $y(0), y(1)$ from recursion, never at $y(-1)$, $y(-2)$ (T3 Q4: $y(0)=1-(-1)-\frac{1}{4}(2)=1.5$).

Repeated root. Needs $n\lambda^n$; for $\lambda=1$ the modes are 1 and n.

$m\ge k$ **in** $h(n)$**.** The mode form starts late; add δ terms.

Geometric off by one. $\sum_0^n$ has $n+1$ terms; $a=b$ gives $(n+1)a^n$, not $0/0$.

No range on a convolution. Attach $u(n)$ or the piecewise ranges; finite windows need several cases.

"Marginally stable" is unstable for BIBO: $|\lambda|=1$, $u(n)$, $(n+1)u(n)$, $\cos(\omega_0 n)u(n)$ are not absolutely summable (ST1 Q4).

ST1 Q4(d) uses the given h, $(2^{-n}+3^{-n})u(n)$, not the answer to (c).

L05

Input phase. $\cos(\omega n+\phi)$ gives $|H|\cos(\omega n+\phi+\theta)$; both phases appear. DC is multiplied by $H(0)$, not 1.

Negative real H**.** Magnitude is positive; add $\pm\pi$ to the phase (T4 Q4).

$H(\omega)$ **needs stability.** $H(z)$ at $z=e^{j\omega}$ only if the ROC contains the unit circle.

L06

Linear via DFT. $N \geq L + M - 1$, not $\max(L, M)$ or $L + M$.

Resolution. Zero padding gives a finer grid, not resolution. $\Delta F = F_s/L$; round N **up** to 2^m (T5 Q5: 120 becomes 128).

Circular index. All DFT shifts and reversals are mod N: $x((-n))_N = x(N - n)$, not $x(-n)$.

L07

No ROC. $X(z)$ without its ROC is half an answer. $a^n u(n)$ and $-a^n u(-n-1)$ have the same $X(z)$.

Reversal ROC. $x(-n)$ inverts **and** swaps the radii (T6 Q1d: $\frac{1}{3} < |z| < 2$ becomes $\frac{1}{2} < |z| < 3$).

L08

Anti-causal sign. $\frac{A}{1-pz^{-1}}$, $|z| < |p|$ gives $-Ap^n u(-n-1)$, not $Ap^n u(-n)$.

Non-monic factors. $(3 - z^{-1})$ must become $3(1 - \frac{1}{3}z^{-1})$ before cover-up (T7 Q1).

Improper fraction. Equal z^{-1} degrees give a $K\delta(n)$ term; K is the ratio of the **highest**-power coefficients (T8 Q5: $0.64/(-1.2) = -0.5333$).

Causal vs inside the circle. Each term's causality comes from the pole's position relative to the **ROC**; the unit circle only decides which ROC is stable.

Unilateral shift. $y(n-2) \to z^{-2}Y + z^{-1}y(-1) + y(-2)$: three terms. No IC terms for a causal input.

L09

Hz vs rad/sample. $\omega = 2\pi F/F_s$ for digital; $\Omega = 2\pi F$ only for a pure analogue question (T9 Q6).

Notch normalisation. Normalise where the question says (T8 Q3: $\omega = \pi$, $z^{-1} = -1$).

Zero on the circle. Notch and comb filters have no stable inverse. Zeros on the circle are not "minimum phase".

L10

No prewarp. Using ω/T or $2\pi F$ in a bilinear design moves the edges.

δ_p meaning. In this course δ_p is the minimum passband gain $10^{-A_p/20}$ (0.8913 for 1 dB), not $1 - 10^{-A_p/20}$.

Rounding N. Always up: 2.06 → 3, 5.29 → 6.

$\cos^{-1}$ in Chebyshev N. Use $\cosh^{-1}$; the arguments exceed 1.

Which Ω_c. Butterworth passband-exact and stopband-exact Ω_c differ (T9 Q5: 0.8140 vs 1.2799). State your choice.

Window index. Use $M - 1$ in the cosine and shift $h_d(n - \alpha)$. Course Hanning, $M = 5$: $\{0, 0.5, 1, 0.5, 0\}$.

Course answers to check against

All values recomputed in Python (numpy, scipy, sympy). ST1 has no official solutions; its values agree with the Test 1 Reference (section 10).

Question	Answer
ST1 Q1(a) ($10e^{-0.01t}\cos(400\pi t)$, $F_s = 100$ Hz)	$T_s = 0.01$ s; $x(n) = 10e^{-0.0001n}$: $10, 9.9990, 9.9980, 9.9970$ (200 Hz aliases to DC)
ST1 Q1(b)	$\{2, \underset{\uparrow}{1}, 0.5, 0, 1\}$, $n = -1..3$
ST1 Q2 ($x(-2..3) = 8, 9, 1, 9, 6, 1$; $x(0) = 1$ checked on the PDF)	(a) $\{8, 9, 1, 9, 14, 10, 1, 9, 6, 1\}$, $n = -3..6$ ($x(0) = 9$). (b) $\{8, 0, 9, 0, \underset{\uparrow}{1}, 0, 9, 0, 6, 0, 1\}$, $n = -4..6$. (c) $\{8, -9, \underset{\uparrow}{1}, -9, 6, -1\}$. (d) $E = 264$
ST1 Q3 (L / TI / C / static)	(a) L, C. (b) L, TI, C. (c) TI only. (d) L, TI. None static
ST1 Q4(a)	$\lambda = e^{\pm j\pi/3}$; modes $\cos(\pi n/3)$, $\sin(\pi n/3)$; $\lvert\lambda\rvert = 1$
ST1 Q4(b)	$\{\underset{\uparrow}{1}, 5.25, 5.3125, 0.328125, -4.917969, -5.229492, \ldots\}$
ST1 Q4(c)	$h(n) = [\cos\frac{\pi n}{3} + 3\sqrt{3}\sin\frac{\pi n}{3}]u(n) = \{1, 5, 4, -1, -5, -4, \ldots\}$; causal, IIR, unstable
ST1 Q4(d)	$(2 \cdot 2^{-n} + 4 \cdot 3^{-n} - 4 \cdot 4^{-n})u(n) = \{2, 1.3333, 0.6944, 0.3356, \ldots\}$
T1 Q1	(a) $T_0 = 2\pi/5$ s; (b), (c), (d) aperiodic; (e) $N = 16$
T1 Q2	Nyquist rate 720 Hz; $x(n) = -2\sin(0.8\pi n)$, $\omega = 0.8\pi$; $y(t) = -2\sin(480\pi t)$
T1 Q3	(a) $L = 128$, 7 bits; (b) $L = 636$, $\log_2 636 = 9.313$, 10 bits
T1 Q4	$\Delta = 1/255 = 0.0039$ V; 160 bits/s; $F_{max} = 10$ Hz
T2 Q1 (static / linear / TI / causal / stable)	(a) S, NL, TI, C, St. (b) D, L, TI, NC, unstable. (c) S, L, TV, C, St. (d) D, L, TV, NC, St. (e) S, NL, TI, C, St. (f) D, L, TV, NC, St
T2 Q2	nonlinear: $x_3(n) = \frac{1}{3}x_2(n-1)$ but $y_3(n) \neq \frac{1}{3}y_2(n-1)$
T2 Q3	$y = 0$ for $n \leq -4$, $n \geq 5$; $\sum_{k=-3}^{n}\alpha^k$ ($-3 \leq n \leq 0$); $\sum_{k=-3}^{0}\alpha^k$ ($n = 1$); $\sum_{k=n-4}^{0}\alpha^k$ ($2 \leq n \leq 4$)
T2 Q4	$h = h_1 * (h_2 - h_3 * h_4) = 2u(n) + 3u(n-1) + 3u(n-2) + 2u(n-3) = \{2, 5, 8, 10, 10, \ldots\}$; causal, unstable
T3 Q1 (L / TI / C)	1: L, TV, NC. 2: L, TV, C. 3: NL, TI, C. 4: L, TV, NC
T3 Q2	no modes; $y = \frac{1}{4}5^{-n}(u[n] + 10u[n-1] + 25u[n-2]) = \{0.25, 0.55, 0.36, 0.072, \ldots\}$
T3 Q3	modes $2^{-n}, 4^{-n}$; $y = (-6 \cdot 2^{-n} - 9 \cdot 4^{-n} + 16 \cdot 3^{-n})u[n]$; $y[2] = -0.28472$
T3 Q4	$\lambda = -\frac{1}{2}$ twice; $y = 1.25(-2)^{-n} + 0.5n(-2)^{-n} + 0.25 \cdot 2^{-n}$, $n \geq 0$ ($y[0..3] = 1.5, -0.75, 0.625, -0.3125$). The printed $y[1]$ line has "2.5" for $y[0]$; it should be 1.5. The result -0.75 is right
T3 Q5	(a) $\{1, 2, 3, 0, 4\}$. (b) $(\frac{1}{3}0.2^n + \frac{2}{3}(-0.4)^n)u[n]$. (c) $(1+n)u[n]$
T3 Q6	(a) $\{2, -3, -3, -5, 6, 5, 10, 12\}$. (b) $(1.2 \cdot 3^n - 0.2 \cdot 2^{-n})u[n] = \{1, 3.5, 10.75, \ldots\}$. (c), (d) as in L04 table (checked numerically)
T3 Q7	$R_{xy} = \{-6, 7, 6, 9, 11, 3, -2, -4\}$, $l = -3..4$ (tutorial definition)
T4 Q2 ($\{-1, 2, \underset{\uparrow}{-3}, 2, -1\}$)	$X(0) = -1$; $\angle X(0) = \pm\pi$; $\int X = -6\pi$; $X(\pi) = -9$; $\int\lvert X\rvert^2 = 38\pi$
T4 Q4 ($y = x(n) + x(n-4)$)	$H = 2e^{-j2\omega}\cos 2\omega$. (i) $3\sin(\frac{\pi}{3}n + \frac{13\pi}{30})$ ($H(\pi/4) = 0$). (ii) $20 + 8.0902\cos(\frac{2\pi}{5}n + \frac{7\pi}{10})$
T5 Q5 ($B = 3$ kHz, $\Delta F \leq 50$ Hz)	$F_s = 6$ kHz; $N \geq 120 \to 128$; $T_0 = 128/6000 = 0.0213$ s

Question	Answer
T6 Q2	$y = -2(\frac{1}{3})^n u(n) + \frac{4}{3}2^n u(-n-1) + \frac{10}{3}(\frac{1}{2})^n u(n)$, $\frac{1}{2} < \lvert z \rvert < 2$
T6 Q3	$h = \{-\frac{1}{2}, \underset{\uparrow}{\frac{3}{4}}, -\frac{1}{2}, -\frac{1}{4}, 0, -\frac{1}{2}, 1\}$; not causal
T7 Q1	$X = \frac{1}{1-2z^{-1}} - \frac{1}{1-\frac{1}{3}z^{-1}}$; ROCs $\lvert z \rvert > 2$, $\frac{1}{3} < \lvert z \rvert < 2$, $\lvert z \rvert < \frac{1}{3}$ (and the empty one)
T7 Q2 (absolutely summable)	$x = -\frac{3}{8}(\frac{1}{3})^n u(n) - \frac{27}{8}3^n u(-n-1)$, $\frac{1}{3} < \lvert z \rvert < 3$
T7 Q3	$H = \frac{1-\frac{1}{2}z^{-1}}{1-\frac{7}{12}z^{-1}+\frac{1}{12}z^{-2}}$; $h = [3(\frac{1}{4})^n - 2(\frac{1}{3})^n]u(n)$; stable
T8 Q2 ($M+1=5$, $F_s = 1$ kHz)	nulls at 200 Hz and 400 Hz
T8 Q3 (50 Hz, $F_s = 500$ Hz, $H(\pi) = 1$)	$\omega_0 = \pi/5$; $b_0 = \frac{1/2}{1+\cos(\pi/5)} = 0.2764$
T8 Q5 stable inverse	$\frac{0.3478}{1-0.8z^{-1}} + \frac{1.1855}{1+1.5z^{-1}} - 0.5333$; $h_I(n) = 0.3478(0.8)^n u(n) - 1.1855(-1.5)^n u(-n-1) - 0.5333\delta(n)$, $0.8 < \lvert z \rvert < 1.5$
T9 Q5 (1 dB, 20 dB, 3/9 kHz, $F_s = 30$ kHz)	$\Omega_p = 0.6498$, $\Omega_s = 2.7528$, $N = 2.0595 \to 3$, $\Omega_c = 1.2799$ (stopband exact); $b = [0.0780, 0.2339, 0.2339, 0.0780]$, $a = [1, -0.7958, 0.5020, -0.0824]$; passband gain -0.074 dB
T9 Q6 ($R = 50$, 80 dB at 100 Hz)	$N = 3.639 \to 4$
T9 Q7 (Chebyshev II, 1 dB, 50 dB, 2/4 kHz, $F_s = 40$ kHz)	$\Omega_p = 0.3168$, $\Omega_s = 0.6498$; course $\epsilon = 0.5087$, $N = 5.2851 \to 6$. Recomputed: $\epsilon = 0.5088$, $N = 5.2929$ (5.2939 with the rounded values). The course 5.2851 could not be reproduced; $N = 6$ either way
T9 Q9 ($M = 5$, $\omega_c = \pi/6$)	rect: $\{0.1378, 0.1592, 0.1667, 0.1592, 0.1378\}$; Hanning: $\{0, 0.0796, 0.1667, 0.0796, 0\}$

DSP · L02–03

Sampling, DT signals, system properties

Cheat sheet · Day 02 · Sun 25 Oct · pack D1

Reference-sheet lines

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```
SAMPLING  f0=F0/Fs, w0=2*pi*F0/Fs. Reduce w0 into (-pi,pi]. Nyquist RATE = 2Fmax.
  Alias: Fa = F0 - k*Fs in (-Fs/2, Fs/2].  sin(-a) = -sin a (sign flips), cos(-a) = c
  Ideal DAC: substitute n = Fs*t AFTER reducing.  cos(2*pi*k*n) = 1 for integer k.
PERIODIC (DT)  f0 = k/N in lowest terms -> N.  Leftover pi -> aperiodic.  Sums: N = L
ADC  D = (xmax-xmin)/(L-1);  b = ceil(log2 L);  bit rate = b*Fs;  |eq| <= D/2;  Fmax
ENERGY/POWER  E = sum|x|^2 (finite -> energy, P=0).  P = lim 1/(2N+1) sum_{-N}^{N}|x
SEQUENCES  u(n+a)-u(n-b) = 1 on [-a, b-1].  x(n-d): shift right.  x(Mn): keep multipl
  x(n/L): zeros between samples.  x(an+b): substitute each n directly.  ALWAYS mark r
PROPERTIES  Static: only x(n).  Linear: x=0 -> y=0, no x^2/|x|/cos x/offset.
  TI: [input delayed, explicit n untouched] vs [every n -> n-d].  Explicit n coeffici
  Causal: no x(n+k), check n<0 too.  Stable: bound |y| by Mx, or u(n) -> infinity dis
  One concrete counterexample disproves a property.
CORRELATION  rxy(l) = sum x(n)y(n-l) (no flip).  rxy(l) = ryx(-l).  rxx even, rxx(0)
  Radar: d_hat = argmax |ryx(l)|;  d = round(2*Fs*r/c).
```

DSP · L04

LCCDE and convolution

Cheat sheet · Day 04 · Tue 27 Oct · pack D2

Reference-sheet lines

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```
LCCDE  sum a_p y(n-p) = sum b_q x(n-q), a0=1, order k
MODES  lambda^k + a1 lambda^(k-1)+...+ak = 0   (y-coeffs ONLY; no y-terms => no modes
  distinct: C1 l1^n + C2 l2^n | double: C1 l^n + C2 n l^n (l=1: C1 + C2 n)
  complex r e^{+-j th}: r^n (A cos th n + B sin th n)
PARTICULAR  same form as x: K a^n (K n a^n if a is a root), K, K1 cos+K2 sin; sub int
TOTAL  y = yh + yp; y(0..k-1) BY RECURSION from ICs; solve for C's; attach u(n); chec
RECURSION  y(n) = -a1 y(n-1) - ... + b0 x(n) + b1 x(n-1)...  (flip y signs)
h(n)  x=delta, zero ICs. FIR: h=b_n. IIR: modes + h(0..k-1) by recursion.
      m>=k: extra delta(n-i) terms, i=0..m-k
CONV  y(n)=sum_k h(k)x(n-k); steps -> limits on k AND range of n
  finite: start = nx+nh, length Nx+Nh-1, check sum y = (sum x)(sum h)
  sum_{k1}^{k2} r^k = (r^k1 - r^(k2+1))/(1-r)
  a^n u * b^n u = (a^(n+1)-b^(n+1))/(a-b) u(n);  a^n u * a^n u = (n+1) a^n u(n)
  check y(0)=h0x0, y(1)=h0x1+h1x0
PROPERTIES  causal: h=0 n<0 | stable: sum|h|<inf (all |lambda|<1) | FIR always stable
  |lambda|=1 or u(n),(n+1)u(n),cos u(n) => UNSTABLE | static: h=K delta(n)
INTERCONNECT  cascade h1*h2 (any order); parallel h1+h2; x*delta(n-d)=x(n-d)
```

DSP · L05

DTFT and its properties

Cheat sheet · Day 06 · Thu 29 Oct · pack D3

Reference-sheet lines

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```
DTFT (L05)  X(w) = sum x(n) e^{-jwn};  x(n) = (1/2pi) ∫_{2pi} X(w) e^{jwn} dw
 converges if sum|x|<inf or sum|x|^2<inf;  X(w+2pi k)=X(w)  -> reduce w to (-pi,pi]
DTFS: x(n)=sum_<N> c_k e^{jk w0 n}, c_k=(1/N) sum_<N> x(n)e^{-jk w0 n}, w0=2pi/N, c_
      (1/N) sum|x|^2 = sum|c_k|^2
SHORTCUTS (mark n=0 at the arrow first!)
 X(0)=sum x    X(pi)=sum (-1)^n x    ∫X dw = 2pi x(0)    ∫X e^{jwn0} dw = 2pi x(n0)
 ∫|X|^2 dw = 2pi sum|x|^2    X'(0) = -j sum n x(n)    angle of negative real = ±pi
PAIRS: delta(n-n0) -> e^{-jwn0};  a^n u(n) -> 1/(1-a e^{-jw}) |a|<1
 u(n)-u(n-M) -> (1-e^{-jwM})/(1-e^{-jw}) = e^{-jw(M-1)/2} sin(Mw/2)/sin(w/2)
 b^n u(-n) -> 1/(1-b^{-1}e^{jw}) |b|>1;   a^n u(n+n0) -> a^{-n0} e^{jwn0}/(1-a e^{-jv
 n a^n u(n) -> a e^{-jw}/(1-a e^{-jw})^2;  (n+1)a^n u(n) -> 1/(1-a e^{-jw})^2
 a^|n| -> (1-a^2)/(1-2a cos w + a^2);  cos(w0 n) -> pi[δ(w-w0)+δ(w+w0)] (period 2pi)
PROPERTIES: x(n-n0)->e^{-jwn0}X ; e^{jw0 n}x->X(w-w0) ; (-1)^n x -> X(w-pi)
 x*(n)->X*(-w); real x: |X| even, phase odd ; x(-n)->X(-w)
 x(n)-x(n-1)->(1-e^{-jw})X ; sum_{-inf}^n x -> X/(1-e^{-jw}) + pi X(0) sum δ(w-2pi k)
 x_(k)(n) -> X(kw) (compress, images) ; n x(n) -> j dX/dw
 x*h -> XH ;  x1 x2 -> (1/2pi) ∫_{2pi} X1(λ)X2(w-λ)dλ (periodic conv.)
SAMPLING: X(w)=(1/T) sum_k Xa((w-2pi k)Fs), w = 2pi F/Fs ; no alias iff Fs > 2 F_M
 recover: ideal LPF gain T, cutoff between W_M and Ws - W_M
METHOD: finite -> term by term; right-sided -> shift index to start at 0, geometric
 series; two-sided -> split n>=0 and n<0; ALWAYS check X(0)=sum x.
```

DSP · L05

Frequency response

Cheat sheet · Day 07 · Fri 30 Oct · pack D4

Reference-sheet lines

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```
FREQUENCY RESPONSE (L05 slides 27-35)
H(w) = sum_n h(n) e^{-jwn}        H(0)=sum h(n)    H(pi)=sum (-1)^n h(n)
LCCDE (all y on left): H(w) = sum_k b_k e^{-jkw} / (1 + sum_k a_k e^{-jkw})
  stable system only (poles inside |z|=1)
e^{jwn}  -> H(w) e^{jwn}
A cos(wn+phi) -> A|H(w)| cos(wn + phi + theta(w))     (sin stays sin)
Sum of sinusoids: do each frequency separately, add. DC: w=0. (-1)^n: w=pi.
Y(w)=H(w)X(w): LTI never creates new frequencies.
  New freq appears -> NOT LTI (nonlinear and/or time-varying).
  No new freq -> proves nothing.
One sinusoid pair gives H at that ONE frequency only.  delta(n) in -> h(n) out.
Symmetric FIR: factor out middle delay: 1+e^{-j2Lw} = 2e^{-jLw}cos(Lw)
  |H| = |real part|; negative real part -> add pi to phase.
1/(1-a e^{-jw}): |H| = 1/sqrt(1-2a cos w + a^2)
GEOMETRY: H = b0 e^{jw(N-M)} prod(e^{jw}-z_k)/prod(e^{jw}-p_k)
  |H| = |b0| prod V_k / prod U_k   (lengths: zero->e^{jw}, pole->e^{jw})
  theta = angle b0 + w(N-M) + sum angle(e^{jw}-z_k) - sum angle(e^{jw}-p_k)
  zero ON circle -> |H|=0 there; pole near circle -> sharp peak
  origin poles/zeros: no effect on |H|, only linear phase
  zero at z=1 kills DC; zero at z=-1 kills w=pi
SKETCH: walk e^{jw} from z=1 (w=0) to z=-1 (w=pi); check 0, pi, pole/zero angles.
```

DSP · L06

DFT and circular convolution

Cheat sheet · Day 09 · Sun 1 Nov · pack D5

Reference-sheet lines

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```
DFT (L06)
X(k) = sum_{n=0}^{N-1} x(n) W_N^{kn},   x(n) = (1/N) sum_k X(k) W_N^{-kn},  W_N = e^-
W_4 = [1 1 1 1; 1 -j -1 j; 1 -1 1 -1; 1 j -1 -j];   W_N^{-1} = (1/N) W_N^*
Checks: X(0)=sum x(n);  x(0)=(1/N) sum X(k);  real x => X(N-k)=X*(k)
Freq sampling: N samples of X(w) <-> x_p(n)=sum_m x(n-mN). Aliasing iff N < L.
   One period: write x in rows of N, add columns.
X(k) = X(w) at w=2pi k/N  (if L <= N).   DFT vs DFS: X(k) = N c_k
FFT: direct N^2 mults; radix-2 (N/2)log2 N mults (O(N log2 N))
Properties (mod N):  x((n-l)) <-> X(k)e^{-j2pi kl/N};  x((-n)) <-> X((-k)) = X(N-k)
   x(n)e^{j2pi ln/N} <-> X((k-l));  x* <-> X*((-k));  x1 (*)N x2 <-> X1 X2
   x1 x2 <-> (1/N) X1 (*)N X2;  Parseval: sum|x|^2 = (1/N) sum|X|^2
Circular reversal of {x0,x1,x2,x3} = {x0,x3,x2,x1}
Circular conv: y(n)=sum_m x1(m) x2((n-m))_N. Circulant: col 0 = x2, shift down + wrap
   Shortcut: linear conv, then fold samples n>=N back by N and add.
Linear conv via DFT: zero-pad to N >= L+M-1, Y=XH, IDFT.
   If N too small: first L+M-1-N samples are corrupted.
Deconvolution: X(k)=Y(k)/H(k) - fails where H(k)=0.
Check any convolution: sum y = (sum x1)(sum x2).
```

DSP · L06

Spectral analysis and block convolution

Cheat sheet · Day 11 · Tue 3 Nov · pack D6

Reference-sheet lines

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```
SPECTRAL ANALYSIS (L06 s29-32)
 Fs >= 2B        Ts = 1/Fs        T0 = L*Ts = L/Fs   (L = real samples)
 Resolution  = 1/T0 = Fs/L        Bin spacing = Fs/N   (same if no zero-pad)
 Bin freq    F_k = k*Fs/N,  w_k = 2*pi*k/N;  k > N/2  ->  -(N-k)*Fs/N
 Real tone F0 -> bins k0 = F0*N/Fs and N-k0; non-integer k0 => leakage
 Design: Fs = 2B; N >= Fs/dF, round UP to 2^m; T0 = N/Fs; quote Fs/N
 Zero-padding: finer grid (interpolation) only, NO new resolution
 Aligned A*cos, rect, N = L: |X(k0)| = A*N/2, all other bins 0
 Hanning: w(n) = 0.5(1 - cos(2*pi*n/(L-1))), 0 <= n < L
   rect : main lobe 2 bins (4pi/L), sidelobe -13 dB
   hann : main lobe ~4 bins (8pi/L), sidelobe -31 dB, gain ~0.5
   => less leakage, worse resolution (budget ~2x the record length)

BLOCK CONVOLUTION (L06 s25-27)   N = L + M - 1, h zero-padded to N once
 OLA: blocks of L + (M-1) zeros; block*h = linear (exactly N long);
      place block i at n = iL; ADD the M-1 overlapping samples
 OLS: block = last M-1 of previous + L new (first block: M-1 zeros);
      circular = linear conv, then fold the last M-1 onto the first M-1;
      DISCARD the first M-1 outputs, keep L
 Blocks: OLA ceil(Lx/L);  OLS ceil((Lx+M-1)/L);  total output Lx+M-1
 T5 Q4 is OLS with M=2: n=0 wrong; fix with +0.5*y(63)
```

z-transform and ROC

Cheat sheet · Day 13 · Thu 5 Nov · pack D7

Reference-sheet lines

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```
z-TRANSFORM (L07)   X(z) = sum x(k) z^-k ;  coeff of z^-n is x(n).  ALWAYS give ROC.
ROC: ring about 0, contains no poles, bounded by poles.
  finite: all z (drop 0 if n>0 samples, drop inf if n<0 samples)
  causal: |z| > outermost pole      anti-causal: |z| < innermost pole
  two-sided: ring between poles (empty -> no transform)
PAIRS: delta(n)->1 ; delta(n-d)->z^-d
  a^n u(n)       -> 1/(1-a z^-1), |z|>|a|
  -a^n u(-n-1)   -> 1/(1-a z^-1), |z|<|a|   (SAME X, different ROC; watch sign)
  n a^n u(n)     -> a z^-1/(1-a z^-1)^2, |z|>|a|
  a^n cos(w0 n)u(n) -> (1-a cos w0 z^-1)/(1-2a cos w0 z^-1+a^2 z^-2), |z|>|a|
PROPERTIES: x(n-d)->z^-d X (x(n+4)->z^4 X) ; z0^n x->X(z/z0), |z0|R
  e^{jw0 n}x -> X(e^{-jw0} z) ; x(-n)->X(1/z), radii inverted & swapped
  x_(k)(n)->X(z^k), R^(1/k) ; n x(n) -> -z dX/dz ; conv -> product, ROC >= RxnRh
  IVT (causal only): x(0) = lim_{z->inf} X(z)
POLES/ZEROS: write in +ve powers of z. z^(N-M) factor -> zeros (N>M) or poles (M>N) a
  X(inf)=0 -> zero at inf ; X(inf)=inf -> pole at inf. #poles = #zeros incl 0, inf.
BLOCKS: cascade H1H2 ; parallel H1±H2 ; feedback G/(1+GF) (minus at adder), G/(1-GF)
  Causal? expand final H(z): any z^{+k} term -> h(-k)≠0 -> NOT causal.
DTFT: X(w) = X(z) at z=e^{jw}, only if ROC contains |z|=1. X(0)=X(z=1), X(pi)=X(z=-1
```

DSP · L08

Inverse z-transform

Cheat sheet · Day 15 · Sat 7 Nov · pack D8

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
INVERSE z (L08)  -- work in w = z^-1, factors as (1 - p w)
A_i = [(1 - p_i w) X]_{w = 1/p_i}      check: sum A_i (+quotient) = X at w=0
A/(1-p z^-1):  |z|>|p| -> A p^n u(n)        |z|<|p| -> -A p^n u(-n-1)
Double pole: 1/(1-pz^-1)^2 -> (n+1)p^n u(n)  |  -(n+1)p^n u(-n-1)
             find A2 by cover-up, then A1 from w = 0
Complex pair p = r e^{j th}:  2|A| r^n cos(th n + angle A) u(n)
                            = 2 r^n [ReA cos th n - ImA sin th n] u(n)
Improper (M >= N in z^-1): divide first; M = N -> constant b_N/a_N
   (ratio of HIGHEST z^-1 coeffs) -> constant*delta(n)
ROC: poles inside ROC's inner edge -> causal terms; outside outer edge -> anti-causal
N distinct pole radii -> N+1 ROCs -> N+1 signals. Stable = ROC contains |z|=1
Power series: coeff of z^-n = x(n). Causal: divide ascending z^-1.
   Left-sided: divide ascending z (highest z^-1 power first)
System ID: H = Y/X (factor, cancel); ROC from "causal"/"stable"; R_y >= R_h n R_x
LCCDE <-> H: y(n) + sum a_k y(n-k) = sum b_k x(n-k)  <->  H = B(z^-1)/(1 + sum a_k z^
   LCCDE alone does not fix the ROC
Causal <-> ROC outside outermost pole (incl. inf). Stable <-> ROC has |z|=1.
Causal+stable <-> all poles |p|<1. Anti-causal+stable <-> all |p|>1.
FIR (MA): M zeros, Mth-order pole at 0, always stable.
All-pole (AR): N poles, Nth-order zero at 0.
```

DSP · L08

Unilateral z-transform and mixed practice

Cheat sheet · Day 16 · Sun 8 Nov · pack D9

Reference-sheet lines

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```
UNILATERAL z-TRANSFORM (L08 s13-14)
X+(z) = sum_{n>=0} x(n) z^-n     (= X(z) if x causal; inverse ALWAYS causal, no ROC c
y(n-1) -> z^-1 Y + y(-1)
y(n-2) -> z^-2 Y + z^-1 y(-1) + y(-2)        <- don't drop the z^-1 y(-1) term
x(n-k) -> z^-k X   for input starting at n=0  (x(-1)=x(-2)=0)
x(n+1) -> zX - z x(0)

2nd order: A(z)Y + I(z) = B(z)X,  I(z) = a1 y(-1) + a2 y(-2) + a2 y(-1) z^-1
  Y_zs = B X / A = H X   (input+system poles)     Y_zi = -I / A  (system poles only)
  y = y_zi + y_zs

PAIRS  p^n u(n) <-> 1/(1-p z^-1)      (n+1)p^n u(n) <-> 1/(1-p z^-1)^2
       n p^n u(n) <-> p z^-1/(1-p z^-1)^2
RESIDUE A_i = (1-p_i z^-1)Y |_{z^-1 = 1/p_i}; double pole: C from squared factor,
        B from z^-1 = 0 (constant terms)
CHECKS  y(0) = Y(z^-1 = 0) = recursion value;  y(1) by recursion
        step steady state = H(1) (stable only)
L04 CHECK  modes + y_p, fit at y(0),y(1) from recursion (NOT at y(-1),y(-2))
           only y_zi may be fitted at n=-1,-2.  L04 constants != y_zi coefficients
KILL A MODE p: numerator of Y = 0 at z^-1 = 1/p  -> solve for unknown IC
ROC (bilateral): causal |z|>max pole; stable <=> ROC contains |z|=1;
                 causal & stable <=> all poles inside unit circle
```

DSP · L09

Digital filters

Cheat sheet · Day 17 · Mon 9 Nov · pack D10

Reference-sheet lines

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```
L09 DIGITAL FILTERS
w0 = 2*pi*F0/Fs  (rad/sample);  F = w*Fs/(2*pi);  Omega = w*Fs
|H(w)| = |b0| * prod(dist to zeros)/prod(dist to poles); poles at 0 don't change |H|
LP: poles near z=1, zeros near z=-1.  HP: reverse.  BP: poles r e^{+-jw0}.
Linear phase: theta = -w*n0 ; tau = -d theta/dw ; symmetric FIR -> n0=(N-1)/2
  phase may jump by +-pi where real amplitude changes sign (still linear phase)
LP->HP: h_HP(n) = (-1)^n h_LP(n) ; H_HP(z)=H_LP(-z) ; flip odd powers of z^-1 ; p ->
FIR design: H(w)=b0+b1 e^{-jw}+b2 e^{-j2w}; null H(w0)=0; DC: sum b=1; pi: b0-b1+b2=1
Notch (FIR):  b0(1 - 2cos(w0) z^-1 + z^-2)
Notch (IIR):  b0(1 - 2cos w0 z^-1 + z^-2)/(1 - 2r cos w0 z^-1 + r^2 z^-2); width ~ 2(
Resonator:    b0(1 - z^-2)/(1 - 2r cos w0 z^-1 + r^2 z^-2)
Comb/MA:      (1/(M+1)) (1 - z^-(M+1))/(1 - z^-1); nulls F_k = k Fs/(M+1)
              |H| = (1/(M+1)) |sin((M+1)w/2)/sin(w/2)|, phase -Mw/2
Allpass:      z^-N A(z^-1)/A(z); poles/zeros reciprocal; 1st order (a+z^-1)/(1+a z^-1
              tau(w) = (1-a^2)/(1+2a cos w + a^2)
Oscillator:   b0/(1 - 2cos w0 z^-1 + z^-2); b0=sin w0 -> h(n)=sin(w0(n+1))u(n)
              y(n) = 2cos w0 y(n-1) - y(n-2) + b0 x(n)
INVERSE: H_I = 1/H ; poles of H_I = zeros of H
  zeros all inside  -> minimum phase    -> stable inverse causal
  zeros all outside -> maximum phase    -> stable inverse anti-causal
  mixed             -> nonminimum phase -> stable inverse non-causal (ring ROC)
  zero ON circle    -> no stable inverse
STABLE INVERSE h_I(n): partial fractions A_i/(1-p_i z^-1), A_i = [(1-p_i z^-1)H_I]_{z
  |p_i|<1 -> A_i p_i^n u(n) ;  |p_i|>1 -> -A_i p_i^n u(-n-1)
  equal degrees -> + K delta(n), K = last num coef / last den coef
  check: sum A_i + K = H_I(z->inf)
```

DSP · L10

IIR design: bilinear transform and Butterworth

Cheat sheet · Day 19 · Wed 11 Nov · pack D11

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
IIR DESIGN BY BILINEAR TRANSFORM (L10)  -- take T = 1 in designs (T cancels)
Map:      s = (2/T)(1 - z^-1)/(1 + z^-1)        s=0 -> z=1,  s=inf -> z=-1
Warping:  Omega = (2/T) tan(w/2)    <->   w = 2 arctan(Omega T / 2)
          w = 2 pi f / Fs   (digital);   Omega = 2 pi f (pure analogue only)
6 STEPS: 1 digital specs wp, ws, dp, ds   2 PREWARP wp, ws -> Wp, Ws
         3 analogue |H(W)|   4 H(s)   5 bilinear -> H(z)   6 CHECK specs
BUTTERWORTH  |H(W)|^2 = 1/(1+(W/Wc)^2N)      Wc = 3-dB point = pole radius
 dp = 10^(-Ap/20) (MIN passband gain)   ds = 10^(-As/20)
 N = log[(10^(As/10)-1)/(10^(Ap/10)-1)] / (2 log(Ws/Wp))     ROUND UP
 Wc (passband exact, slide 17) = Wp / (10^(Ap/10)-1)^(1/2N)
 Wc (stopband exact, T9 Q5)    = Ws / (10^(As/10)-1)^(1/2N)   -> say which!
 LHP poles s_k = Wc exp(j(pi/2 + pi(2k+1)/(2N))), k=0..N-1;  H(s)=Wc^N/prod(s-s_k)
 N=2: s^2+1.4142Wc s+Wc^2      N=3: (s+Wc)(s^2+Wc s+Wc^2)
 N=4: (s^2+0.7654Wc s+Wc^2)(s^2+1.8478Wc s+Wc^2)
 Pair at angle th: s^2 - 2Wc cos(th) s + Wc^2
FACTOR METHOD (K = 2/T):
 s+a      -> [(K+a) + (a-K) z^-1] / (1+z^-1)
 s^2+bs+c -> [(K^2+bK+c) + 2(c-K^2) z^-1 + (K^2-bK+c) z^-2] / (1+z^-1)^2
 numerator Wc^N (1+z^-1)^N ; divide by den constant so a0 = 1
 diff eq: y(n) = -a1 y(n-1) - a2 y(n-2) ... + b0 x(n) + b1 x(n-1) ...
CHECK: gain_dB = -10 log10[1+(W/Wc)^2N] at Wp and Ws (= digital gain at wp, ws)
ORDER FROM POLE RADIUS R: Wc = R; N = log(10^(A/10)-1)/(2 log(W1/Wc)), W1 = 2 pi f
IMPULSE INVARIANCE (low priority): h(n)=T ha(nT), pole p -> e^(pT); aliases
```

DSP · L10

Chebyshev and FIR design

Cheat sheet · Day 20 · Thu 12 Nov · pack D12

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
CHEBYSHEV (I and II, same eps and N)          [L10 s24, s27]
  dp = 10^(-Rp/20), ds = 10^(-As/20)
  eps = sqrt(1/dp^2 - 1)          (1 dB -> 0.5088, 0.5 dB -> 0.3493)
  N = acosh( sqrt(1-ds^2)/(eps*ds) ) / acosh(Ws/Wp)   ROUND UP
      acosh x = ln(x + sqrt(x^2-1)); slides print "cos^-1"
  Digital: Wi = 2 tan(wi/2) (T=1) BEFORE taking Ws/Wp
  Type I : equiripple passband, |H(0)|=1 (N odd) or dp (N even), all-pole
  Type II: equiripple stopband, zeros at W = Ws/cos((2k+1)pi/2N)
CHEB I POLES  beta = ((sqrt(1+eps^2)+1)/eps)^(1/N)
  r1 = Wp(beta^2+1)/(2beta) [jW axis], r2 = Wp(beta^2-1)/(2beta) [sigma]
  phi_k = 90 + (2k+1)180/(2N) deg ; p_k = r2 cos phi_k + j r1 sin phi_k
  K = prod(-p_k) (N odd), /sqrt(1+eps^2) (N even)
FAMILIES  order: Elliptic < Cheb I = Cheb II < Butter < Bessel
          phase: Bessel best ... Elliptic worst
LINEAR PHASE FIR  alpha=(M-1)/2, group delay = alpha
  sym: H = e^{-j alpha w} A(w)   antisym: H = e^{-j(alpha w - pi/2)} A(w)
  I sym/odd: any | II sym/even: zero z=-1 (no HP)
  III anti/odd: h(alpha)=0, zeros z=+-1 (BP only) | IV anti/even: zero z=1 (no LP)
  Fill: compare ends -> sym or anti; mirror about alpha
WINDOW METHOD  hd(n) = sin(wc n)/(pi n), hd(0)=wc/pi
  course sinc x = sin x / x  -> numpy: (wc/pi)*np.sinc(wc*n/pi)
  h(n) = hd(n - alpha) w(n),  n = 0..M-1
  Rect 1 | Bartlett 1-2|n-alpha|/(M-1) | Hanning 0.5(1-cos(2pi n/(M-1)))
  Hamming 0.54-0.46cos(2pi n/(M-1)) | Blackman 0.42-0.5cos(.)+0.08cos(2.)
  M=5: Hanning=Bartlett={0,.5,1,.5,0}; Hamming={.08,.54,1,.54,.08}
  stopband: rect -21, Bart -26, Hanning -44, Hamm -53, Black -75 dB
  A(w) = h(alpha) + 2 sum h(n) cos((alpha-n)w)  -> check at 0, wc, pi
FREQ TRANSFORMS (analogue)  HP: s -> Wp Wp'/s
  BP: s -> Wp (s^2+Wl Wu)/(s(Wu-Wl)), centre sqrt(Wl Wu), order doubles
  Digital HP: z^-1 -> -(z^-1+a)/(1+a z^-1), a = -cos((wp+wp')/2)/cos((wp-wp')/2)
  Check: new edge has the same gain as the prototype edge
```